Prices vs. Production Restrictions: China's Coal Industry*

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Abstract

Governments often use command-and-control production restrictions to reduce pollution. Unlike taxes, these restrictions need not reduce output in order of producer efficiency and can create misallocation. We develop a criterion to compare the welfare effects of production restrictions and taxes under mild assumptions on the damage function. We study China's 2016 coal reform, which reduced coal output to support industry profitability while reducing pollution. Using an empirical model of inter-provincial coal demand and supply, we show that the observed policy was less efficient than a tax but can be rationalized by policy preference for producer surplus.

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1 Introduction

Regulators often use price- and quantity-based policies to mitigate externalities. Economists have long advocated for market-based price or quantity instruments, such as Pigouvian taxes (Pigou, 1920; Baumol, 1972) or cap-and-trade systems (Crocker, 1966; Dales, 1968; Montgomery, 1972), to internalize externalities efficiently. Although resource taxation is common, sophisticated cap-and-trade markets are often infeasible due to high transaction costs (Nie, 2012) or weak institutional capacity (Bell and Russell, 2002; Blackman and Harrington, 2000), especially in developing countries. In these cases, regulators may resort to directly restricting production, such as factory shutdowns or moratoria on resource extraction (Blackman, Li and Liu, 2018).

However, comparing the welfare effects of these policy instruments can be challenging. First, demand, supply or the scale of externality may be hard to measure precisely. For example, estimates of the Social Cost of Carbon vary by orders of magnitude depending on discount rates and tipping point assumptions (Pindyck, 2013). Second, policy makers may care about some welfare outcomes, such as consumer or producer surplus, more than others. The government's welfare function thus could place different weights on different welfare components.

We study a major reform of China's coal industry launched in 2016 and compare the welfare effects of the reform with a tax. Policy choices in this setting are consequential. As the world's largest producer and consumer of coal, China faces acute tensions between energy security, environmental consequences, and economic stability. It has been widely reported that the reliance on coal undermines China's energy transition and global efforts to mitigate climate change (McGrath, 2019; WSJ, 2022; Bloomberg, 2022). A number of papers have documented the severe air-quality and health consequences associated with coal (Almond et al., 2009; Chen et al., 2013; Ebenstein et al., 2017). At the same time, coal mining provides significant employment opportunities and local government revenues, especially in major producing provinces dominated by state-owned enterprises (Wright, 2012; Hervé-Mignucci et al., 2015). Policies affecting coal output therefore have large economic and distributional consequences (Clark and Zhang, 2022; Yuan et al., 2025).

As part of a multi-year effort to reduce coal consumption and pollution while preserving employment and profitability of the industry, the Chinese government in April 2016 adopted a production restriction policy, ordering small mines to close and large mines to reduce output. As provinces implemented these measures, coal prices surged: our data show that the average coal price in the second half of 2016 rose by 31.0% to RMB 539 ($77.4)/ton, from RMB 411 ($60)/ton in the same period of 2015. The government continued to close mines and strictly monitor production to reduce production above the approved capacity at least through 2018.

From a welfare perspective, why might the government prefer this policy to taxation? Using a local approximation based on (Weitzman, 1974), we first show that it is theoretically ambiguous whether a pollution tax or a production restriction achieves higher social welfare. Second, we show how the preferred policy may change when the government places a greater weight on certain objectives, such as firm profits rather than consumer surplus. Third, we propose a criterion for comparing the two policies when researchers observe either the benefits or the costs of abatement, but not both. The key idea is based on stochastic dominance. Suppose, for example, that we can measure abatement costs under both policies and that higher quantities generate weakly higher pollution damages. If one policy has higher abatement costs (more lost economic surplus) and also induces a stochastically dominating distribution of quantity (more pollution), then the policy must imply a lower social welfare.

To analyze the demand and supply in the coal market, we combine the policy intervention with a comprehensive dataset on coal production and consumption. Our main data set includes monthly prices at the province level and shipment quantities from producers to end users both within and across provinces. We validate the shipment data with the production and consumption data from other sources.

We next formulate a model that incorporates key institutional features of China's massive and partially regulated coal market. First, coal procurement in the electricity and other sectors has undergone significant market reforms since 2013, essentially allowing the market to set prices that equilibrate demand with supply. Second, the shares of coal shipped from one province to another are highly stable over time, despite varying price dispersions across provinces. Third, large coal producers and buyers often operate under long-term quantity contracts. These contracts at the time often promised a delivery quantity but allowed prices to adjust based on spot market prices. In our model, the market is competitive. We specify a demand function and a supply function for each province. The local price in each province is determined by the province's demand and the total supply from different provinces. The quantity supplied by a province is determined by a weighted average of prices in provinces it sells to. The effects of the policy are modeled as a common threshold change in supply costs once production approaches capacity.

We estimate the demand and supply of coal across provinces. To identify demand, we construct price instruments using variations in the policy start dates and policy compliance levels across provinces that shift supply. The key identifying assumption is that the policy exclusively affects supply and is not correlated with the demand unobservables. We also consider a variety of alternative instruments. Similarly, we estimate supply using demand shifters. Our preferred threshold estimates suggest that the policy significantly steepens aggregate marginal costs once production approaches capacity.

Given the estimated equilibrium model of coal demand and supply, we conduct a series of counterfactual simulations to quantify the effects of enforcing the production restriction. First, we find that removing the restrictions would increase annual production by 668 million tons and decrease the price by RMB 145/ton. Removing the restrictions would raise economic surplus by RMB 259 billion, mostly through consumer gains from lower coal prices, but it would also increase pollution damages by RMB 309 billion. Measured welfare is therefore RMB 50 billion higher under the observed production restriction than under no policy.

Second, we compare the policy with an efficient quota or a cap-and-trade system to understand its distortionary effects. Our supply estimates suggest that the policy increases aggregate marginal costs especially near approved capacity, generating inefficient reductions of quantities both within and across provinces. Holding fixed each province's observed quantity, removing the within-province distortion would raise producer surplus by RMB 136 billion. Efficiently reallocating quantity reductions across provinces would increase the total producer-surplus gain to RMB 212 billion.

We then use the estimates to rank the production restriction policy against a tax. We first consider a welfare function that consists of consumer and producer surpluses, tax revenue, and avoided pollution damages that we can measure. We find that an optimal tax of RMB 417/ton would raise measured welfare by RMB 261 billion relative to the observed policy.

Next, we consider a more general welfare function, which includes an unknown pollution damage function that increases in total quantity. This damage can reflect social cost of carbon, but more generally captures any social costs that we do not quantify but rise with aggregate production. Using the proposed criterion, we show that the tax above can achieve a higher welfare for any such damage function.

Finally, we rationalize the policy choice with heterogeneous policy preferences for producer surplus and avoided pollution damages. We find that, when the weight on avoided pollution damages is one, the observed policy would outperform any tax policy if policymakers value producer surplus 40.8% more than the other components of welfare. The intuition is that the weighted loss of producer surplus under a tax can be so large that tax revenue cannot offset it, whereas production restrictions can directly increase producer surplus. We also find that a greater weight for avoided pollution damages must be paired with an even greater preference for producer surplus for the government to prefer the observed policy to a tax, because taxes tend to reduce quantities more for the same producer surplus.

Related Literature and Contribution

We first contribute to a literature that uses observed policy choices to understand the political or social preferences. In trade, Goldberg and Maggi (1999) estimate the government's relative weight on aggregate welfare versus political contributions. Subsequent work examines related political-economy models of trade protection (Gawande and Bandyopadhyay, 2000; Mitra, Thomakos and Ulubaşoğlu, 2002; McCalman, 2004; Matschke and Sherlund, 2006; Gawande, Krishna and Olarreaga, 2009; Francois and Nelson, 2014; Adão et al., 2026). Related work estimates social welfare weights that rationalize observed tax-benefit systems (Bourguignon and Spadaro, 2012; Bargain et al., 2014; Hendren, 2020) and infers political motives from infrastructure choices, including rail in California and highways in China (Fajgelbaum et al., 2024; Alder and Kondo, 2019). Relative to this literature, which typically infers preferences from the intensity of a given policy instrument (such as tariff rates or tax levels), we use the government's choice between instruments, the observed production restriction vs. tax, to bound the relative weights on components of the welfare function.

We build on the classic framework of Weitzman (1974)[1] to compare taxation with production restriction. Weitzman (1974) studies the choice between price and quantity instruments under uncertainty, while subsequent work compares taxes with permits, standards, command-and-control rules, and hybrid policies (Adar and Griffin, 1976; Roberts and Spence, 1976; Koenig, 1985; Helfand, 1991; Dickie and Trandel, 1996). Our comparison differs in two respects. First, we relate the welfare comparison to policy preferences and show how they interact. Second, our counterfactual analysis simulates the market equilibrium under alternative policies, going beyond a local approximation. We provide an equilibrium framework to compare the instruments, even when the benefits of abatement are not fully measured. Our approach based on estimated demand and supply models also differs from prior work comparing policy instruments, which uses integrated assessment models (Pizer, 2002; Muller and Mendelsohn, 2009) or calibrated theoretical models (Newell and Pizer, 2003; Fell, MacKenzie and Pizer, 2012).

Our paper also contributes to the empirical literature on measuring the economic effects of environmental regulations.[2] A number of papers focus on productivity impacts (e.g., Jaffe et al., 1995; Berman and Bui, 2001; Greenstone, 2002; Greenstone, List and Syverson, 2012;

[1] For the large literature related to Weitzman's framework, see, for example, Williams III (2002), Hepburn (2006), Weitzman (2020), and Stavins and Wagner (2022).

[2] More broadly, we note that there is a growing literature on the empirical evaluation of industrial policy (Kline and Moretti, 2014; Aghion et al., 2015; Alder, Shao and Zilibotti, 2016; Juhász, 2018; Kalouptsidi, 2018; Lane, 2022; Miravete, Moral and Thurk, 2018; Lashkaripour and Lugovskyy, 2018; Criscuolo et al., 2019; Rotemberg, 2019; Barwick, Kalouptsidi and Zahur, 2019; Giorcelli, 2019; Yi, Lawell and Thome, 2019; Hanlon, 2020; Bai et al., 2020; Fan and Zou, 2021; Giorcelli and Li, 2021; Guo and Xiao, 2022; Aldy, Gerarden and Sweeney, 2023).

He, Wang and Zhang, 2020). Using estimated equilibrium models, Ryan (2012) and Fowlie, Reguant and Ryan (2016) emphasize the roles of market power. Chen et al. (2025) study how conglomerates reallocate internal production. In this paper, we develop and estimate a new competitive equilibrium model for China's coal sector.

We also note that there have been few equilibrium analyses of China's coal industry in economics despite its enormous importance to China's economy and the global environment. One reason might be the lack of granular and recent data.[3] For example, Zhou et al. (2019) study the evolution of coal mining firms' productivity, and Zheng (2024) studies the effects of buyer market power on productivity and safety, both using a production function approach based on manufacturing census data from before 2007. In our paper, we combine a detailed dataset, the institutional knowledge of the market, and the industry reform to address the endogeneity problems in the estimation of coal demand and supply in our equilibrium model. Our estimates of demand and supply elasticities may be of independent interest for other researchers studying China's coal industry.

In the rest of the paper, we first describe the main institutional features of China's coal market in Section 2. Then in Section 3, we illustrate the welfare comparison between tax and production restriction in a conceptual model. Section 4 lists the various datasets used in the analysis. Sections 5 and 6 present the demand and supply models and their estimation. We use the model to analyze the reform of China's coal industry and how the observed policy choice reveals policy preferences in Section 7. Section 8 concludes.

2 Empirical Background

2.1 Coal Types, Quality, and Mining Technology in China[4]

Bituminous coal accounts for more than 70% of Chinese coal production and anthracite for another 23%.[5] High-purity bituminous coal and roughly half of anthracite are used for coking, which produces coke for steel-making. The remaining output of these two coal types

[3]Existing studies that estimate or calibrate the coal demand or supply typically use aggregate annual data (see, for example, Burke and Liao, 2015; Shi, Rioux and Galkin, 2018; Teng, Burke and Liao, 2019). A number of papers have studied the coal-mining industry in other countries, with a focus on safety and productivity (Sider, 1983; Gowrisankaran et al., 2015), technology adoption (Rubens, 2022), labor market competition (Delabastita and Rubens, 2022; Demirer and Rubens, 2025), and the decline of the US coal industry (Watson, Lange and Linn, 2023).

[4]This section is based on Nathaniel Aden, David Fridley and Nina Zheng (2009) and National Energy Report (2020).

[5]Coal is usually classified by carbon content by weight. The carbon content is over 86% for anthracite and between 45% and 86% for bituminous coal. The remaining coal falls into lower-carbon categories, such as subbituminous coal and lignite (American Geosciences Institute, n.d.).

is used mainly by power plants for electricity production. Lower-purity coal is used primarily in cement production and heating.

Coal purity and extraction methods differ markedly across regions. Shanxi and Inner Mongolia hold the largest reserves and are the largest coal-producing provinces, together accounting for more than a quarter of national output.[6] Shanxi also produces the highest-quality coal, with an average heat content, defined as the total energy released per kilogram after complete combustion, of about 6,242 kcal/kg, compared with a national average of 5,350 kcal/kg and a US average of 5,600 kcal/kg. More than 90% of Chinese mines are pithead mines, where coal is extracted from deep underground. The average mine depth is 456 meters. This extraction method is more costly than open-pit mining more common in countries such as Australia and India, and in regions such as the western United States.

2.2 The Formation of the Coal Market

For decades, the Chinese government tightly controlled both coal supply and coal demand through planned prices and quantities. A partial liberalization in 1993 allowed coal to be traded, but preserved special contracts under which large state-owned power plants could buy coal at planned quantities and prices (Yang et al., 2018). These special contract prices were typically below prevailing market prices. When contracted volumes fell short of electricity demand, power plants purchased the balance on the spot market. Subsequent reforms culminated in the abolition of the special contracts in 2013 (State Council, 2013). These reforms reshaped the procurement, operating, and investment decisions of power plants, which began to respond to coal-price fluctuations more like profit-maximizing firms (Xu and Chen, 2006; Ma, 2011; Liu, Margaritis and Zhang, 2013; Zhao and Ma, 2013; Gao and Van Biesebroeck, 2014; Eisenberg, 2019, 2024).

2.2.1 Method of Sales

Delivery Contracts. During our sample period, coal buyers and producers transact through both long-term contracts (annual or multi-year agreements) and the spot market. The major buyers are coal-fired power plants, municipal central-heating systems,[7] construction firms, steelmakers, and fertilizer manufacturers. A long-term contract is signed directly between a coal producer and an end user, rather than a trading intermediary. Provincial governments

[6] Appendix Figure I.1 shows the distribution of coal reserves.

[7] Across northern China, most residential apartments, government facilities, and schools receive heating through municipal centralized systems that primarily burn coal to generate heat. This infrastructure consists of large coal-fired boilers that heat water, which is then distributed through an extensive network of insulated pipes to various buildings (Chen et al., 2013).

frequently help broker these matches, particularly in large coal-producing provinces, which bore political responsibility for supplying power plants and heating systems elsewhere in the country (NDRC, 2021*b*). The contracts nominally specify an annual delivery quantity but allow prices to adjust quarterly, or even monthly, with the spot market (e.g., Shanghai Securities News, 2014, 2015; ICBC, 2016). Realized deliveries can therefore depart from stated contract terms (International Energy Network, 2016*a*).

Monthly Price Discovery. In practice, buyers and sellers in long-term contracts set prices using public monthly price indices. The most widely used indices during and after our study period are regional average transaction prices, net of transportation costs, for coal sold to power plants (NDRC, 2016*b*, 2017). Buyers and sellers update contract prices using the initial signing prices, which can reflect coal quality as well as projected supply and demand, and the previous month's average delivery prices published by coal trade associations and port authorities, which may pool both contract and spot transactions.[8]

Contract Compliance. Contemporaneous press reports indicate that almost no contracts were honored at the contracted price between 2014 and 2016. Major miners sold at contract prices through the first two months of 2014 but cut prices once market prices began to falter (Li, 2014; Zhao, 2014, 2015). [9] Compliance remained poor in 2015 and 2016 (Bie, 2015; Wang, 2015).

Compliance improved markedly in 2017 for three reasons. First, the industry reform described below raised prices. Second, the new contracts specified a price range that was easier to honor amid spot-market fluctuations. Third, the government introduced penalties for power plants that failed to take their full contracted volume, including exclusion from the wholesale electricity market (Gao, 2017). These measures compressed the spread between

[8]International Energy Network (2016*b*) provides an example of how to reset the monthly price using the average price of coal sold to power plants (BSPI) and the average price of coal for other uses (OPI) around the Bohai Gulf, both published by Qinhuangdao Ocean Shipping Coal Trading Market Co. Given the contract price p, the indices p_t^{BSPI} and p_t^{OPI} in month t, and the heat content of coal h measured in kcal/kg, the new price in month $t+1$ is adjusted to

$$0.5p + \left(0.25p_t^{\text{BSPI}} + 0.25p_t^{\text{OPI}}\right) \frac{h}{5,500\text{kcal/kg}}.$$

The government advises buyers and sellers to explicitly specify this formula (dynamic price updates linear in average coal prices) in their long-term contracts. Contract prices, price indices, and assigned weights could differ across firms and years.

[9]Hu (2014) reports that contracts at the beginning of 2014 covered 1.9 billion tons of coal. Our data suggest that the first two months' quantity typically accounts for 15% of the annual level. A back-of-the-envelope calculation implies that the contract-compliant share in the early months of 2014 would be about 7.3% of total coal production (3.87 billion tons in our data).

spot and contract prices (China National Coal Association, 2018). Later reports (Zhang, Hong and Wu, 2020) put contract-compliant volume at an average of 2 billion tons per year over 2016–2020, roughly 57% of the 2017 quantity.

2.3 Economic Policies

2.3.1 Coal Supply Reform

The 2016 five-year plan from the National Development and Reform Commission (NDRC), China's macroeconomic management agency, set out a broad agenda for restructuring the coal-mining industry. Specifically, the government explicitly called for raising the industry's profitability, limiting coal production, and reducing the industry's environmental impact.[10] Based on this plan, the State Council, China's chief administrative body, ordered in February 2016 that coal mines cut their operating schedule from 330 to 276 working days per year and that small mines be closed. We refer to these orders as the **supply reduction policy**.[11]

Implementation timing and compliance varied across provinces and coal mine ownership types. Shanxi, historically the largest coal-producing province, moved first, in April 2016. Other leading coal-producing provinces such as Shaanxi and Inner Mongolia followed in May. Large state-owned mines faced more stringent monitoring and cut output by more (Appendix I.2). Coal prices surged in the second half of 2016, both relative to earlier in the year and relative to the same months of 2015. In October the NDRC suspended the 276-working-day rule out of concern for meeting the (highly inelastic) winter heating and power demand (Lai, 2016; Shi, Rioux and Galkin, 2018). Nonetheless, mine closures and strict production monitoring continued through 2017 and 2018. The policy was eased only gradually beginning in 2019 (NDRC, 2019).

2.3.2 Resource Taxation

China introduced a resource tax on coal production in 2014. The central government set a rate band of 2 to 10% on the sales revenues of the raw coal extracted,[12] within which provincial governments choose a specific rate subject to the central government's approval (STA, 2014). Major provinces such as Shanxi set the rate at 8%, while others chose lower rates (Qin, Zhang and Xie, 2020).

[10] In Appendix A, we provide snapshots of the original document and its translation.

[11] The Chinese press often referred to this policy as "capacity reduction policy," reflecting the closure of small mines. Given that the policy also reduced the output of operating mines, we use the term "supply reduction" as a compromise.

[12] This revenue levy does not additionally distinguish between different grades of the coal, but allows for reduced rates on coal extracted in mines near depletion.

Policymakers did not make this tax the principal instrument of the 2016 reform for several reasons. First, the tax was designed to lighten firms' overall burden and to fund provincial spending on coal-related environmental damage as part of a broader fiscal reform (Xinhua, 2014), which replaced a patchwork of fees on coal production with a single levy. Raising the rate to restrict output would have cut against that purpose. Second, fiscal reform moves slowly, whereas the central government viewed cutting coal capacity and output as urgent and better suited to fast, targeted administrative mandates (Li, 2016; People's Daily, 2016). Third, the central government allows provincial governments to retain all resource-tax revenue as an incentive to invest in environmental protection (Hu, 2017), leaving little room to redistribute the proceeds to coal consumers or producers.

2.3.3 Emission Trading System

China's carbon market for coal-fired power was introduced through regional emissions-trading pilots and then consolidated into the national Emissions Trading System (ETS) for the power sector. The institution was developed over a long horizon: seven regional pilots launched in 2013–14 and, after several years of operation, covered only about 7% of national CO_2 emissions (Almond and Zhang, 2021), while the national program did not begin trading until 2021 and initially applied only to the power sector (Goulder et al., 2022). The ETS adopts a tradable performance standard rather than cap-and-trade (Ministry of Ecology and Environment, 2020; Goulder et al., forthcoming): allowances are allocated through fuel- and technology-specific emissions-intensity benchmarks, with pre-allocation tied to lagged generation and final allocations adjusted to match the realized output.[13] These features incentivize power plants to reduce emissions intensity rather than to reduce generation or retire coal capacity. Both the institutional design and the protracted rollout of the ETS were therefore poorly aligned with the 2016 reform's objective of reducing coal output.

2.3.4 Other Economic Policies

Two further sets of policies during our sample period are worth noting. First, the government expanded credit and infrastructure spending to offset the decline in housing and stock market prices in 2015 and 2016 (Brandt et al., 2020). Many infrastructure projects, such as the national high-speed rail system, experienced rapid expansion in subsequent years (Reuters, 2016). Second, the government began closing small or unproductive plants in other indus-

[13] Allowances in 2019–2020 were first pre-allocated using 70% of 2018 electricity/heat supply, but after 2019–2020 emissions data were verified, the final quota was set according to the generator's actual 2019–2020 electricity/heat supply; if the final approved quota differed from the pre-allocation, excess allowances were withdrawn or missing allowances were added.

tries.[14] These closures cut state-owned steel capacity by 10% in 2016 (Lu, 2016). However, we expect the plant closures to have a small effect on coal demand given that the output share of closed plants is likely low. Between 2016 and 2018, coal demand remained strong as the main consuming sectors—thermal power, coking, steel and cement—either expanded output or maintained historically high production levels, against the backdrop of continued economic growth.[15]

3 Conceptual Framework

We compare a tax and a production restriction using a local approximation approach based on Weitzman (1974). We model a production restriction as an increase in the slope of the supply, because a policy like the one in Section 2 removes production capacity, thus limiting supply's response to prices. Unlike a quota, the policy does not fix output. We show that, as a result, uncertainties about demand, supply and externality can all matter for the welfare ranking of policy instruments.

3.1 Setup

Social welfare from a regulated activity with output q is $W = B(q) - C(q)$, where B and C capture the social benefits and costs. The optimal quantity $\hat{q}$ satisfies $B' = C'$. To analyze the welfare effects of policy instruments that shift quantities near $\hat{q}$, we follow Weitzman (1974) and take a quadratic approximation approach. Let $x \equiv q - \hat{q}$ denote the deviation from $\hat{q}$. The welfare difference is approximately

$$\Delta W \equiv W(q) - W(\hat{q}) \approx (\beta - \alpha)\, x - \tfrac{D}{2}\, x^2, \tag{1}$$

where α and β are mean-zero, independent shocks to the marginal social cost and benefit at $\hat{q}$ with variances σ_α^2 and σ_β^2, and $D \equiv C'' - B''$. We assume that the marginal benefit of abatement diminishes ($B'' < 0$) and the marginal cost increases ($C'' > 0$), and thus $D > 0$. In an application where output generates negative externalities, α captures the shocks to demand and supply, and β captures the shocks to marginal damage from pollution.

[14] It is of separate interest that the government, as opposed to the market, needs to order the shutdown of the plants. A plausible explanation is that many plants are supported by the local governments to provide employment despite being unprofitable. To facilitate the shutdown of coal and steel plants, China's Ministry of Finance created a fund of RMB 100 billion ($15.3 billion) to provide relief to the laid-off employees (Lu, 2016).

[15] We plot the total output in Appendix C.1.

3.2 Policy Instruments

A policy maker chooses an instrument before shocks realize to maximize expected welfare $E[\Delta W]$. The instruments differ in how quantity adjusts after shocks.

Tax. The regulator sets a per-unit tax t on output that equates the marginal social benefit and cost of abatement: $t = C'(\hat{q}) = B'(\hat{q})$. Given the realized shock α, the quantity adjusts so that the marginal cost matches t, and the realized deviation of quantity from $\hat{q}$ is:

$$x_{\text{tax}} = -\alpha / C''. \tag{2}$$

Quota. The regulator fixes output at $\bar{q}$. Maximizing $E[\Delta W]$ over $\bar{q}$ yields $\bar{q} = \hat{q}$: $x_{\text{quota}} = 0$.

Production Restriction. For a given restriction that raises the marginal-cost slope from C'' to $C'' + \kappa$ ($\kappa > 0$), quantity adjusts to equate the marginal benefit to the steeper marginal cost after shocks:

$$x_{\text{restriction}} = \frac{\beta - \alpha}{D + \kappa}. \tag{3}$$

As $\kappa \to \infty$, $x_R \to 0$ and the restriction converges to a quota.

3.3 Welfare Ranking

Substituting each instrument's x into (1) and taking expectations yields the following comparisons.

Tax vs. Quota (Weitzman, 1974)

$$E[W_{\text{tax}} - W_{\text{quota}}] = \frac{C'' + B''}{2(C'')^2}\,\sigma_\alpha^2. \tag{4}$$

A tax dominates when the marginal cost of production increases steeply relative to the rate at which the marginal benefit of abatement diminishes ($C'' + B'' > 0$); a quota dominates when the marginal benefit of abatement diminishes sufficiently steeply ($C'' + B'' < 0$). Only cost-side uncertainty σ_α^2 enters because the quota eliminates all quantity variation.

Tax vs. Production Restriction

$$E[W_{\text{tax}} - W_{\text{restriction}}] = \underbrace{\frac{C'' + B''}{2(C'')^2}\,\sigma_\alpha^2}_{\text{tax vs. quota}} - \underbrace{\frac{D + 2\kappa}{2(D+\kappa)^2}\left(\sigma_\alpha^2 + \sigma_\beta^2\right)}_{\text{restriction vs. quota}}. \tag{5}$$

Equation (5) differs from (4) in two ways. First, the uncertainty about marginal benefit of abatement σ_β^2 now enters: because the production restriction does not fix quantity, the adjustment to optimal output depends on deviation of the marginal benefit of abatement, and its variance affects welfare. Second, the comparison decomposes into the Weitzman advantage of a tax over a quota ($E[W_{\text{tax}} - W_{\text{quota}}]$) minus the advantage of the restriction over a quota ($E[W_{\text{restriction}} - W_{\text{quota}}]$). The second term is always positive because, locally, the restriction dominates a pure quota for allowing partial quantity adjustment to shocks. As $\kappa \to \infty$, the second term vanishes and (5) reduces to (4).

We note that if the policy maker has no uncertainty over the cost of abatement, which is the loss of economic surplus when quantity is reduced ($\sigma_\alpha = 0$), the above implies that restriction would unambiguously dominate a tax. In reality, restrictions would often shift up not only the marginal but also the inframarginal supply curve, creating more welfare losses the local approximation here does not capture. We explore a global comparison between tax and restriction in Section 3.5.

3.4 Welfare Weights

We now consider a welfare function that weighs components differently. We specialize the analysis to the empirical context in Section 2, where the government prioritizes improving producer surplus and reducing pollution. We focus on comparing tax and production restriction.

Under a tax $t > 0$, the government with weight w_{PS} on producer surplus and w_{damage} on pollution damages evaluates the tax at

$$\tilde{W}_{\text{tax}} = CS + w_{\text{PS}}(R - C - tq) + tq - w_{\text{damage}}\, L, \tag{6}$$

where CS, R, tq and L are consumer surplus, revenue, tax revenue and the pollution damage. A production restriction generates no tax revenue:

$$\tilde{W}_{\text{restriction}} = CS + w_{\text{PS}}(R - C) - w_{\text{damage}}\, L. \tag{7}$$

We now sign the difference between the objective values under the tax and production restriction for higher w_{PS} or w_{damage}.

Producer weight. Under the optimal tax and restriction, the envelope theorem implies that

$$\frac{\partial\, E[\tilde{W}_{\text{tax}} - \tilde{W}_{\text{restriction}}]}{\partial w_{\text{PS}}} = E[(R{-}C{-}tq)_{\text{tax}} - (R{-}C)_{\text{restriction}}] \approx -t\hat{q} \;<\; 0. \tag{8}$$

The derivative shows that production restriction is more likely to be the better instrument when the government values producer surplus. The intuition is that the tax revenue $t\hat{q}$ extracted from producer surplus (weighted by $w_{\text{PS}} > 1$) is devalued when converted to tax revenue (weighted by one).

Damage weight.

$$\frac{\partial\, E[\tilde{W}_{\text{tax}} - \tilde{W}_{\text{restriction}}]}{\partial w_{\text{damage}}} = \underbrace{-\frac{\sigma_\beta^2}{D+\kappa}}_{\text{damage-shock response}} - \frac{L''}{2}\left(\frac{\sigma_\alpha^2}{(C'')^2} - \frac{\sigma_\alpha^2 + \sigma_\beta^2}{(D+\kappa)^2}\right). \tag{9}$$

The first term shows that the restriction (Eq. (3)) has a greater first-order effect on output compared with a tax, lowering expected damage through less output. The second term may mitigate this effect when the tax generates less output variance as the restriction ($\sigma_\alpha^2/(C'')^2 < (\sigma_\alpha^2 + \sigma_\beta^2)/(D+\kappa)^2$), which limits the chance of catastrophic damage under tax when the damage function is sufficiently convex.

Together, a greater weight on producer surplus unambiguously rationalizes the production restriction, while whether a greater weight on pollution damages does so depends on the relative magnitudes of cost-side and damage uncertainty. Our empirical exercise estimates demand and supply for coal in China and asks which combinations of welfare weights can explain the observed policy choice. In doing so, we also account for the welfare losses on inframarginal units of production that arise when the restriction shifts the supply curve.

3.5 Comparing Instruments Using Stochastic Dominance

The welfare ranking in Section 3.3 relies on a local approximation and requires curvature assumptions on benefits and costs. We now present a complementary, global comparison for settings in which the researcher can compute the equilibrium distribution of output and economic surplus under each policy, which can be, for example, based on estimated demand and supply functions, but does not know the curvature of the pollution damage function L.

The key idea is to use stochastic dominance to identify when a tax improves upon a production restriction for any increasing damage function. A classical result in decision theory (Quirk and Saposnik, 1962; Hadar and Russell, 1969) provides a tool for comparing $E(L(Q))$ across policies without knowing the functional form of L: for any increasing function L and two random variables Q_1 and Q_2, $E(L(Q_1)) \leq E(L(Q_2))$ if their distributions satisfy $F_{Q_1}(q) \geq F_{Q_2}(q)$ for all q (first-order stochastic dominance).

This yields the following procedure to assess whether a tax can improve upon a production restriction for the welfare function

$$W^\star = \underbrace{E(CS + R - C)}_{\mathcal{E}} - E(L(Q))$$

1. Compute the expected economic surpluses $\mathcal{E}_{\text{restriction}}$ and $\mathcal{E}_{\text{tax}}$ under the production restriction and the tax.

2. Compute the respective distributions of quantities.

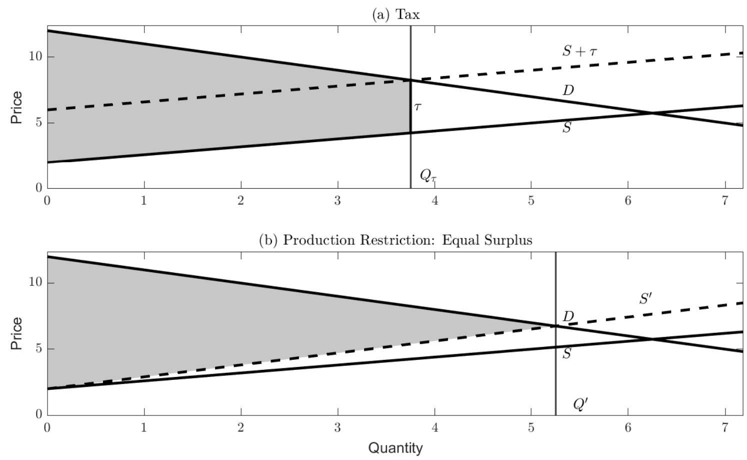
3. If (1) the distribution of output under the tax is stochastically dominated by that under the production restriction, which implies $E(L(Q_{\text{tax}})) \leq E(L(Q_{\text{restriction}}))$ for any increasing L, and (2) $\mathcal{E}_{\text{tax}} \geq \mathcal{E}_{\text{restriction}}$, the tax achieves higher $W^\star$.

Why a surplus-improving tax can have lower output. A production restriction that constrains low-cost producers makes the aggregate supply curve steeper. A tax, by contrast, preserves the original supply curve and reallocates production toward lower-cost producers. Figure 1 illustrates this point: at any given level of aggregate output, a tax generates more economic surplus (the shaded area between demand and supply) than the production restriction with a steeper supply S'. Equivalently, to achieve the same surplus, the equilibrium quantity under a tax (Q_τ) is lower than under the restriction (Q'). When the tax can generate at least as much surplus with lower quantities across the support of the shocks, the output distribution under the tax is stochastically dominated, and the tax improves welfare for any increasing damage function.

Global vs. local comparison. This stochastic dominance approach goes beyond the local comparison in Section 3.3. The local approximation assesses how a tax and a production restriction differ in their response to shocks near a common benchmark quantity. When a production restriction is large enough to rotate a significant portion of the supply curve, it misallocates production and creates large inframarginal losses. These losses can be sufficiently large that a tax can achieve higher expected surplus at lower quantities. No additional convexity restriction is needed for welfare comparison.

A dual formulation. The same logic extends to comparing policies when marginal benefits of abatement are relatively well measured but the costs of abatement are not. Suppose a pollution-control policy reduces market quantity below the no-policy equilibrium. Under the conditions that private surplus increases and cost decreases in quantity, economic surplus is then increasing in quantity up to the no-policy equilibrium level. Therefore, between two policies that reduce pollution damages by the same amount, the policy that induces a stochastically dominating distribution of market quantity, thus preserving higher output, delivers higher welfare.

Figure 1: Tax Generates More Surplus at Lower Quantity



Panel (a): under a tax τ, the equilibrium quantity is Q_τ and the shaded area is the economic surplus. Panel (b): a production restriction rotates supply from S to S'; to generate the same surplus, the quantity Q' must be higher.

4 Data

4.1 Data Sources

Our primary dataset comes from the China Coal Transportation and Distribution Association (CCTDA), a major trade group in the coal industry. The data include monthly shipment quantities of coal between provinces from 2012 to 2016. The tonnage data combine raw coal, washed coal and other derivative products with high coal content (such as coke and briquette).[16] The shipment data include shipments within a province and to other provinces.

We then compile additional data from other sources. We collect province-level mining capacity data from 2014 to 2018 from the National Energy Administration and National Mine

[16] Consistent with the standard NBS practice, the data account for transportation of the tonnage of final end products from the raw coal producers to the end users. More specifically, if 1 ton of raw coal was extracted in province A, transported to province B for washing (removing impurities) with quantity reduced to $x < 1$, and burned at power plants in province C, the data would only include x in transportation from A to C and not count the $A - B$ transportation. We also note that not all raw coal needs washing, and many coal mines are integrated with washing facilities and need not transport coal to a third province for processing. Our data closely match the NBS aggregate production and consumption. See Appendix B for more details.

Safety Administration. Any capacity change is approved every 6 months.[17] We also obtain province-month production data in 2014–17 from CCTDA and 2018–2019 from National Bureau of Statistics (NBS). The production data reflect the total tonnage of excavated raw coal. We obtain monthly production data for key state-owned (SOE) coal mines from CCTDA, which are the largest coal mines in each province.[18] The monthly coal imports, measured in total tonnage, from outside China to each province between 2014 and 2017 are from the firm International Coal.[19] We collect monthly provincial coal prices in 2014–2019 from NDRC.[20] To study the impact of coal consumption on air pollution, we collect air quality data (hourly concentration measures of pollutants such as PM2.5 and PM10) at the monitoring station level from the Ministry of Environmental Protection.

We do not have data on monthly province-level coal consumption. To construct coal consumption at the province-month level, we impute the quantities based on (1) shipment data from CCTDA, (2) monthly output of major industries using coal, and (3) annual province-level coal consumption, both from NBS. Details of this imputation are in Appendix C.

To verify the consistency of data from different sources, we compare the CCTDA's shipment data out of a province with the production data from NBS in Appendix B. Our shipment data are slightly lower (by about 4% to 8%) than the NBS statistics. At a more granular level, the same pattern holds consistently across provinces.

Data Aggregation

In the rest of the paper, we use the total weights of coal produced and consumed to analyze supply and demand. We do not further differentiate among coal types within the quantity data. Given our research questions, several potential concerns about the data may warrant discussion. Importantly, to what extent can a single demand function capture the composite demand for a variety of coal and coal products combined in our quantity data? What is the appropriate price for this composite demand? We note that in China, the demand for different coal products is likely highly correlated. The widespread use of coal blending techniques

[17] In principle, the annual production should not exceed the approved level, and monthly production should not exceed 10% of the 1/12 of the annual level.

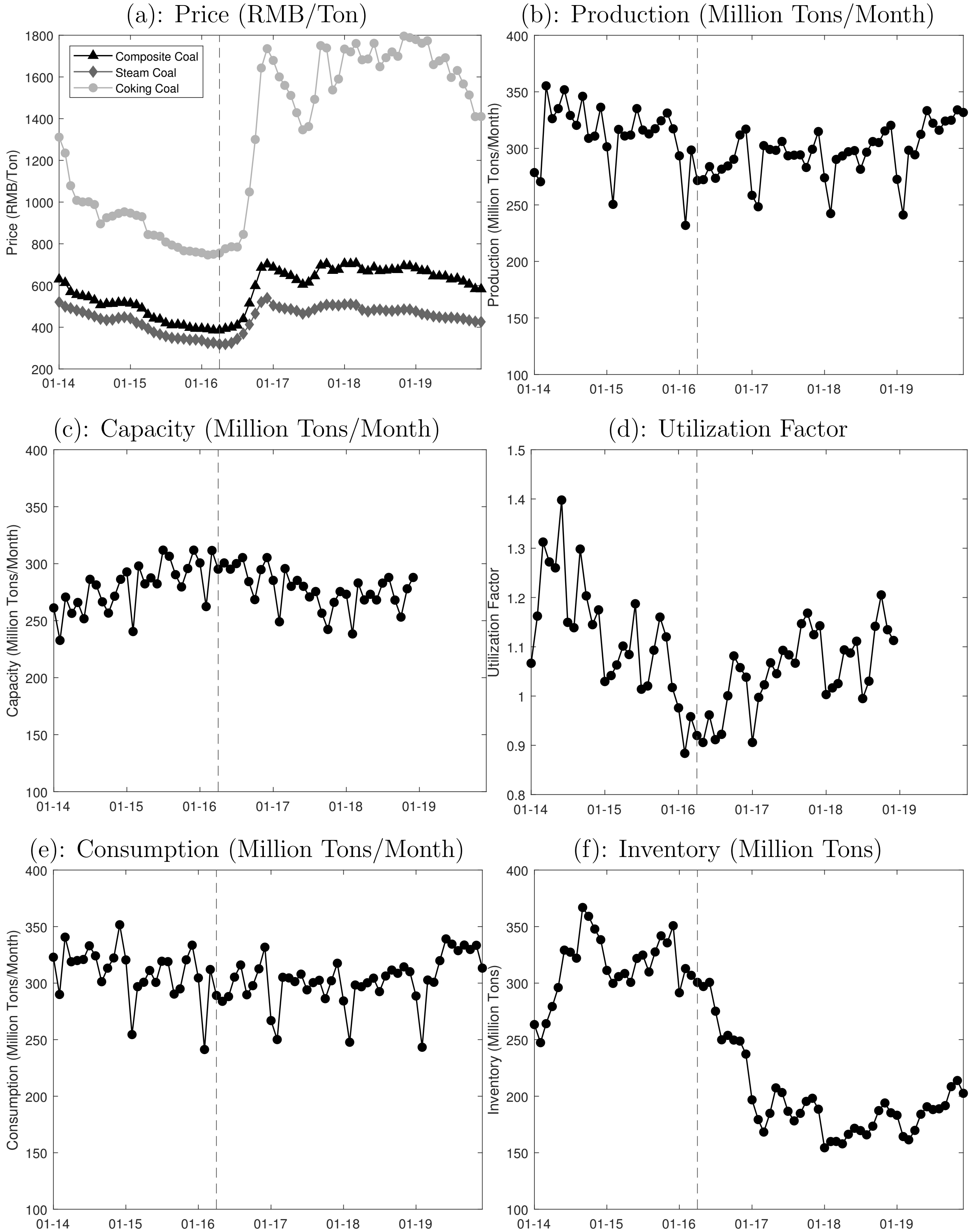
[18] NDRC designates an SOE coal mining firm as a "key SOE" based on a number of criteria, which include capacity, reserves, mining technology and coal quality (whether the coal is mainly for electricity production).

[19] The firm provides the data through the website https://mcoal.in-en.com/. We do not consider China's exports, which account for less than 0.2% of coal production during our sample period.

[20] The publicly available coal price data are based on coal used for electricity generation and heating. This type of coal, which includes a variety of coals with different heat contents and purity, is collectively referred to as "steam coal" or "thermal coal." We collect monthly province-specific prices of steam coal using the China Steam Coal Price Index from the Price Monitoring Center of NDRC. The index is based on data collected from 1,600 firms that rely on coal as the main fuel source, including major coal-fired power plants and heating facilities. Prices of coal used for coking are based on prices in Ganqimaodu in Inner Mongolia, a major hub of coking coal distribution.

enables power plants to substitute across different coal types in the event of shortages. Other end users such as heating facilities are likely even less sensitive to coal quality. As we will show next, thermal and coking coal prices are highly correlated. Therefore, we construct an average price for each province and month based on the two price series weighted by their consumption shares.

Figure 2: Time-Series Data of the Coal Market



Notes: Panel (a) plots the prices of steam coal, coking coal and the composite price, (b) plots the national monthly production, (c) plots the monthly national capacity, which is adjusted from the capacity that changes every 6 months for the number of working days in each month (our capacity data end at the end of 2018), (d) is the utilization factor, defined as national monthly production divided by capacity, (e) is the national monthly consumption, and (f) is the national inventory. The vertical line indicates the start of the supply reduction policy.

4.2 Data Summary

4.2.1 National Time Series

We plot the time series of the national coal prices, production, capacity, utilization factor, inventory and consumption in Figure 2. Coal prices shown in (a) increased significantly for both coking and steam coal as the policy started in Shanxi in April 2016. The prices remained higher throughout 2017 and 2018 and gradually decreased in 2019 after the policy pressure eased. We plot the monthly production in (b). The production in 2016 was noticeably lower than similar periods both before and after the policy.

The supply squeeze was a result of two aspects of the policy. First, the government closed coal mines and suspended new approvals, resulting in a net decrease in capacity as shown in (c). Second, the campaign to reduce over-production above approved capacity levels directly limits output. In (d), we plot the utilization factor, defined as the production-to-capacity ratio. This ratio in 2017 and 2018 (post-policy) is lower than or comparable to 2014–15, despite significantly higher prices in 2017 and 2018. The utilization factor is also lower in 2016 than in 2017–2018, reflecting more stringent enforcement in 2016 (the 276-working-day policy).

In (e) and (f), we find that the market went through a transition in 2016. The figure (e) shows that consumption has a more muted response in 2016 than production. Excess demand over production corresponds with an over 40% decrease in inventory (Figure 2.(f)) during the rapid price increase in 2016.[21] The inventory stabilized at a lower level in 2017 and 2018, when production increased above 300 million tons per month, matching the level of consumption.

[21] Monthly foreign imports also increased by about 10 million tons per month in 2016 and remained at about 20 million tons per month post-policy.

Table 1: Summary Statistics

	Mean	SD	p25	Median	p75
Panel A. Coal-Consuming Provinces					
Consumption (Million Tons/Month/Province)	10.16	7.63	4.62	7.47	13.30
Price (RMB/Ton)	532.01	133.87	430.21	523.03	615.45
Wind Generation (TWh/Month/Province)	0.53	0.73	0.06	0.29	0.70
Cooling Degree Days (°C/Month/Province)	17.77	38.16	0	0	11.12
N=1440; based on 30 coal-consuming provinces in 2014–2017					
Panel B. Coal-Producing Provinces					
Quantity (Million Tons/Month/Province)					
Supply	11.30	19.91	1.20	3.61	9.98
Capacity	11.21	17.85	1.59	5.18	12.40
N=1200; based on 25 coal-producing provinces in 2014–2017					
Panel C. Foreign Import					
Quantity (Million Tons/Month)	18.12	4.97	14.00	17.55	21.82
N=48; 2014–2017					

Notes: Panel A shows summary statistics for 30 coal-consuming provinces. The demand consists of shipments from Chinese provinces and foreign imports. Coal consumption is aggregated from consumption in all sectors using coal for each province (Appendix C). The price is the quantity-weighted average of thermal and coking coal prices plus a calibrated transportation cost (Appendix E) based on coal origins. We also use monthly wind generation and cooling degree days in the analysis. Panel B reports summary statistics for 25 coal-producing provinces. Supply is the sum of all shipments from that province each month. Capacity is aggregated from the mine-level data from NEA. Panel C reports total imports.

4.2.2 Summary Statistics

For the main analysis of the paper, we focus on the years from 2014 to 2017.[22] Table 1 summarizes our main data set. There are 30 coal-consuming provinces and 25 coal-producing provinces. All coal-producing provinces also consume coal. The reported prices include transportation costs based on coal origins, which account for approximately 17.5% of the composite price.[23]

[22] CCTDA's cross-province shipment data are available through 2016, and its production data are available through 2017. We cross-validate both datasets using NBS statistics. For subsequent years, NBS reports province-level monthly production, except for January and February.

[23] Appendix E details how we calibrate the transportation costs.

Table 2: Mean and Standard Deviation of Cross-province Shipment Shares, 2014–2016

	Destination (Consuming) Province									
Origin (Producing) Province	Inner Mongolia	Shanxi	Shandong	Hebei	Jiangsu	Shaanxi	Henan	Liaoning	Guizhou	Xinjiang
Inner Mongolia	0.392	0.011	0.049	0.017	0.064	-	0.002	0.082	-	-
	[0.014]	[0.002]	[0.008]	[0.003]	[0.010]	-	[<0.001]	[0.003]	-	-
Shanxi	0.006	0.298	0.104	0.231	0.103	0.002	0.055	0.017	-	0.001
	[0.002]	[0.013]	[0.007]	[0.029]	[0.012]	[0.001]	[0.012]	[0.002]	-	[0.001]
Shaanxi	0.004	0.148	0.086	0.039	0.101	0.411	0.059	-	0.001	-
	[0.001]	[0.020]	[0.049]	[0.010]	[0.027]	[0.070]	[0.007]	-	[0.001]	-
Guizhou	-	-	-	-	-	-	-	-	0.740	-
	-	-	-	-	-	-	-	-	[0.025]	-
Shandong	-	-	0.857	0.012	0.033	-	0.008	0.002	-	-
	-	-	[0.016]	[0.005]	[<0.001]	-	[0.001]	[<0.001]	-	-

Notes: The table reports the shipment shares of the five largest coal-producing provinces to the ten largest coal-consuming provinces, averaged over 36 months between 2014 and 2016. The share is calculated as a month's shipment from i to j divided by the total shipment from i in the month. The standard deviations are reported in square brackets.

4.2.3 Cross-province Shipment Shares Are Stable Over Time

We show that the share of a province's coal production transported to another province is highly stable over time. This important empirical fact will guide the modeling decisions in the model. In Table 2, we report the mean and standard deviation (in square brackets) of the shipment shares from province i to province j, defined as $\frac{Q_{ijt}}{Q_{it}}$, where Q_{ijt} is the tonnage of coal produced in i and transported to j, and Q_{it} is the total production in i in month t. We list the five largest coal-producing provinces on the row and the ten largest coal-consuming provinces across columns. The standard deviation for the majority of province pairs is negligible relative to the mean, indicating a high degree of stability in shipment shares over time.

There are two possible reasons for this stability despite substantial variation in coal prices over time. First, the government and railway operators allocate railway capacity when coal buyers and sellers enter into long-term contracts. The rail capacity allocations create rigidity within the duration of long-term contracts, as any significant rerouting of shipment across provinces could create cascading disruptions in the interconnected rail network. This constraint incentivizes coal buyers to switch between suppliers within a given province even when coal buyers procure coal from outside their long-term contracts. Second, although power plants can substitute across different grades of coal to some extent through coal-blending technology, they still tend to prefer similar coals when purchasing on the spot market, because coal blending can add to the plants' operating costs (Zhao, 2021). Since mines within the same province are more likely to produce similar grades of coal, they may supply to the same set of power plants, consistent with stability across multiple years.

We also find support for these structural reasons in explaining the rigidity of shipment shares. Between 2014 and 2016, the log difference between the highest and lowest coal prices across provinces varies between 0.62 and 0.94.[24] If shipment shares could change flexibly in response to price differentials, we would expect to observe strategic shipping behaviors to exploit the arbitrage opportunities, keeping price dispersion relatively constant. Instead, the observed persistence of price differentials points to a lack of such strategic behaviors. Our model below thus focuses on production decisions as opposed to shipment destination choices.

[24]The log difference of 25th–75th price levels ranges from 0.12 to 0.35.

5 Coal Demand

We specify coal consumption in province j and month t as

$$\ln q_{jt}\left(p_{jt}\right) = \alpha^{D} \ln p_{jt} + \beta_X X_{jt} + FE_j^{\text{demand}} + FE_{m(t)}^{\text{demand}} + \varepsilon_{jt} \tag{10}$$

where the quantity q_{jt} corresponds to the consumption of coal in province j, and p_{jt} represents the composite coal price in the province.[25] The parameter α^D has the straightforward interpretation of price elasticity. The covariates include province j's one-month lagged cooling degree days (CDD) and wind generation, which affect the local electricity residual demand for coal power plants, and an indicator for 2017 to capture government intervention to shore up long-term contract compliance on the demand side. We also include fixed effects for the province and the month of the year. The term ε_{jt} captures the unobserved demand shock.

We treat wind generation as largely exogenous to the unobserved shocks ε_{jt} for two reasons. First, wind power build-out is primarily driven by centrally planned investment that targets areas rich in wind resources (State Council, 2014). Second, NDRC directs local governments and the power grid to guarantee priority dispatch and full purchase of wind generation (NDRC, 2016*a*). Given these policies, we assume wind production to be driven by an exogenous long-term trend plus local and short-run weather conditions, which generates exogenous variations in residual demand for coal generation.

Static Demand Assumption. In theory, the coal purchase decisions can be dynamic, where coal users can use existing inventory to smooth consumption when prices change (Jha, 2023). Our demand function is static, because coal consumers do not store large quantities of coal in our context. Specifically, NDRC requires coal power plants to keep enough stock for at least 7 days of operation, but explicitly caps inventories at no more than 12 days in peak seasons (NDRC, 2021*a*). A historically high value in 2024 is just 27 days (CCTV News, 2024). Major steel plants keep a similar level of coal inventory (BOCI, 2017). This practice contrasts sharply with the norm in the US, where over 90% of coal power plants keep enough coal to generate electricity for over 60 days (EIA, 2023). We also note that, although there may still be arbitrageurs with sizable capacity, such as dedicated coal traders storing coal at ports, the model in (10) focuses on coal consumption by end users instead of shipment. In our counterfactual analysis, we focus on the "steady state" of the market, where demand matches the level of consumption in the markets before 2016 or in 2017, and the inventory

[25] We note that there may exist potential differences between spot prices and the re-adjusted prices facing buyers and sellers on long-term contracts. Because we do not have contract prices or order-level prices, we use the spot prices as a first-order approximation for sales in a province.

level was roughly stable over time (Figures 2.(e) and (f)).

Long-term Contracts. Another concern is that long-term contracts may reduce demand for spot-market coal. Section 2 shows that this is largely not the case for 2014-2016 when delivery prices and quantities significantly deviated from contracts: buyers frequently purchased from the spot market when market prices were low, and sellers sold to the market demand when prices were high. In 2017, government measures both improved contract compliance while also bringing the contract prices more in line with spot prices, reducing the distinction between demand for contracted and spot market coal. In addition to including a 2017 indicator in demand, we further explore the robustness of our demand estimates in Appendix G where we assume that contracts proportionally reduce the demand for spot market coal.

5.1 Identification

The equilibrium price may be correlated with the unobservable ε_{jt}. We construct shift-share instrumental variables that exploit the timing and exposure differences of provinces to the production restrictions as documented in Section 2.3.1.

Instrumental Variables (IV)

There are three potentially useful variations:

1. The policy started first in Shanxi in April 2016, followed by other provinces in May. Shanxi is also one of the provinces that experienced the largest decrease in output.
2. The policy imposes stricter monitoring programs on the production of state-owned enterprises (SOEs), and the share of production by SOEs differs across provinces.
3. The policy closed small coal mines, and the cutoff thresholds differ across provinces.

In each province, the variations in policy timing and intensity shift the local supply through the largely fixed shipment shares (Section 4.2.3). Importantly, because the demand function (10) already accounts for policy effects through the 2017 indicator, our excluded instruments are based on purely supply-side variations in 2016, when the policy rolls out across provinces at different times and enforcement intensity differs from 2017 (the 276-working-day policy in 2016). Specifically, we use χ_{it} to denote whether the policy has applied to province i's supply in month t, and for each province j in month t, we define (1) a policy IV

$$Z_{jt}^{\text{Policy}} = \sum_i w_{ij}^{2012-2013} \chi_{it}, \tag{11}$$

where $w_{ij}^{2012-2013}$ is the coal shipped from i to j divided by the total shipment to j, averaged across months from 2012 and 2013 (before the estimation sample), which captures j's historical dependence on coal from i.[26] We also define (2) an SOE IV,

$$Z_{jt}^{\text{SOE}} = \sum_i o_i w_{ij}^{2012-2013} \chi_{it}, \tag{12}$$

where o_i is the output share[27] of key SOEs in province i, also averaged across months from 2012 and 2013. In the above, the interactions between o_i and $w_{ij}^{2012-2013}$ proxy the exposure differences to the supply policy shocks. Finally, we define (3) a capacity IV,

$$Z_{jt}^{\text{Capacity}} = \sum_i K_i^{\text{Small}} w_{ij}^{2012-2013} \chi_{it}, \tag{13}$$

and K_i^{Small} is province i's capacity in January 2014 below the province's cutoff thresholds for closure during the policy in 2016.[28]

IV Validity

Our IVs exploit the independence between demand shocks and the shipment shares' interaction with χ_{it} conditional on province fixed effects and month-of-the-year fixed effects, which is the key assumption for the validity of Bartik-type shift-share instruments (Goldsmith-Pinkham, Sorkin and Swift, 2020). This assumption is supported by the policy's implementation pattern: the earliest and largest effects occurred in Shanxi, and SOE production faced tighter restrictions, because policymakers prioritized large coal mines to achieve rapid output reductions. (Appendix A), as opposed to correlations with demand shocks. In particular, the two largest provinces' SOEs significantly reduced their production in 2016 (Appendix I.2).

One potential concern with the exclusion restriction of the shift-share IVs is a correlation between policy start times and other economic policies. We note that the 2015-2016 economic stimulus plan is a nationwide policy that generally eases the borrowing constraints for transportation projects, such as the national network of high-speed rails (HSR). A corre-

[26]We find that the shares of shipment from a province, w_{ij}, just like the shares of shipment to a province in Section 4.2.3, are stable over time during our sample. We argue in Section 4.2.3 that structural reasons not due to changing prices are likely responsible for the rigidity.

[27]Ideally, we should use capacity shares to construct the IV because there can be potential serial correlations in unobservables. We do not have capacity data for SOE mines. The main assumption is that any serial correlation of production between 2013 and 2016 is negligible.

[28]Following the official policy document, the capacity cutoffs for "small" coal mines to be closed are 0.3 million tons/year for major coal-producing provinces: Shanxi, Inner Mongolia, Shaanxi, and Ningxia, 0.15 million tons/year for Hebei, Liaoning, Jilin, Heilongjiang, Jiangsu, Anhui, Shandong, Henan, Gansu, Qinghai, and Xinjiang, and 0.09 million tons/year for the remaining provinces.

lation with our instruments may arise when the first province to enact the policy, Shanxi, is also a major supplier to provinces with more infrastructure projects in April that would use more coal. Appendix D lists major active infrastructure construction projects in April and May of 2016. The large volumes of concrete and steel used in these projects could generate substantial demand for coal and coke. We find no clear correlation between the locations of these projects and Shanxi's main destination provinces—Shanxi, Hebei, Shandong, and Jiangsu—which together account for over 70% of Shanxi's output (Table 2).

Another concern is that policy timing may be endogenous if coal buyers anticipated the policy and stockpiled coal in advance. Policy guidance and common industry practices discussed previously show that major end users of coal do not have significant storage capacity. National coal inventory data from NBS in Figure 2 show that the inventory actually declined before the policy, suggesting that the policy is likely a surprise.

5.2 Estimation

To understand the strengths of the IVs and the underlying variations they capture, we report the first-stage results in Table 3. Columns (1)–(3) regress the natural log of prices on each instrument in the previous section. We find that the results are intuitive. Shanxi's early policy implementation raises the prices in provinces more exposed to its supply. Similarly, prices are higher in provinces where a greater mix of supply historically comes from SOEs or small mines, and given the intensity of the policy, particularly Shanxi's SOEs and small mines. The F statistics indicate that the policy and SOE IVs are considerably stronger.

Table 4 reports the second-stage results based on the instruments in Table 3. Both tables report province-cluster wild-bootstrap standard errors. We use the estimated elasticity of -0.635 based on the policy and SOE IVs in column (5) as our primary specification. Notably, the stronger IVs in columns (2), (3) and (5) all imply similar demand elasticities. Although the estimate based on the weaker IV of small mine capacity (column (4)) is larger, the estimate in our primary specification is not rejected at the 95% confidence level. The IV estimates are considerably different from the positive OLS estimate in column (1). In comparison, prior work (Burke and Liao, 2015) has estimated the elasticity for coal demand to be between -0.3 and -0.7. Finally, we note that our results are robust to using the different IV estimates.

Table 3: Demand Estimation: First-Stage

	(1) Policy	(2) SOE	(3) Small-Mine Capacity	(4) Policy+SOE
Z_{jt}^{Policy}	0.075***			0.017
	(0.014)			(0.030)
Z_{jt}^{SOE}		0.147***		0.117**
		(0.023)		(0.050)
Z_{jt}^{Capacity}			0.086***	
			(0.033)	
Wind Generation (TWh)	-0.067***	-0.066***	-0.056***	-0.067***
	(0.024)	(0.025)	(0.022)	(0.025)
CDD/1000	0.260***	0.249***	0.335***	0.249***
	(0.081)	(0.087)	(0.101)	(0.084)
Province FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
Kleibergen-Paap F Statistic	24.91	37.02	5.93	19.23
N	1440	1440	1440	1440

Notes: The dependent variable is log delivered coal price. Province and month fixed effects are included in all specifications. Standard errors, reported in parentheses, are clustered at the province level using a wild bootstrap. *, **, and *** indicate significance at the 10, 5, and 1 percent levels.

Table 4: Demand Estimation

	OLS	(2) Policy	(3) SOE	(4) Small-Mine Capacity	(5) Policy+SOE
$\ln p_{jt}$	0.115***	-0.696***	-0.619***	-1.258***	-0.635***
	(0.031)	(0.212)	(0.197)	(0.487)	(0.204)
Wind Generation (TWh)	-0.039**	-0.081***	-0.077***	-0.110***	-0.078***
	(0.016)	(0.027)	(0.024)	(0.040)	(0.023)
CDD/1000	0.465**	0.742***	0.716***	0.934***	0.722***
	(0.225)	(0.238)	(0.228)	(0.296)	(0.237)
Province FE	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES
N	1440	1440	1440	1440	1440

Notes: The dependent variable is log coal consumption. Province and month fixed effects are included in all specifications. Standard errors, reported in parentheses, are clustered at the province level using a wild bootstrap. *, **, and *** indicate significance at the 10, 5, and 1 percent levels.

6 Coal Supply

Our supply model incorporates two key institutional features of China's coal industry. First, the coal market is not concentrated. In Appendix I.3, we plot the combined output shares of the three largest coal-mining firms (based on the output in the key SOE production data) in each of the top five coal-producing provinces. No single firm has more than 30% of the output share in a province, and the share of the third largest firm is often below 11%. Second, as we show in Section 5.1, the share of supply from one province to another is highly stable over time.

Motivated by these two observations, we specify the aggregate marginal cost function at the province level as

$$\ln \mathrm{mc}_{it} = \alpha^S \frac{q_{it}}{K_{it}} + FE_i^{\text{supply}} + FE_{m(t)}^{\text{supply}} + \xi_{it}, \tag{14}$$

where K_{it} is the capacity in province i in month t, which is taken as given in the data, and we include fixed effects at the province and month level. We assume that, in equilibrium, the marginal cost matches the effective price $P_{it} = \sum_j s_{ij} (p_{jt} - \tau_{ij})$, where s_{ij} is the share of coal shipped from i to j in the total shipment from i,[29] p_{jt} is the price in the destination province j, and τ_{ij} is the transportation cost calibrated following the procedure in Appendix E.[30]

Behavioral Foundation

The marginal cost function (14) implies the supply curve $\frac{Q_{it}(P_{it})}{K_{it}} = \gamma \ln P_{it} + \widetilde{FE}_i^{\text{supply}} + \widetilde{FE}_{m(t)}^{\text{supply}} + \tilde{\xi}_{it}$, where $\gamma = \dfrac{1}{\alpha^S}$, with fixed effects and shocks also appropriately transformed. The supply function can be microfounded as the aggregate decisions of many atomistic, competitive mines. Suppose each producer in province i sells a share s_{ij} to province j. The cost function is given by $c(q; \theta)$, where the cost parameter θ differs across mines and follows the distribution Θ_i. The total mass of the mines is $\mathcal{M}$. Slightly abusing notation, we use $a_{\theta t}$ to denote this producer's output decision, which satisfies the first-order condition $P_{it} = c'(a_{\theta t}; \theta)$. Under the assumptions that the marginal cost function c' is strictly monotonic in $a_{\theta t}$ for any θ, we can then invert the production decision as $a_{\theta t} = c'^{-1}(P_{it}; \theta)$. Integrating both sides with respect to $\Theta_i(\theta)$ and multiplying the total mass of the mines $\mathcal{M}$ yield an aggregate supply curve as a function of P_{it}, and appropriate choices of c and Θ_i would give the specific exponential form.

[29] We use the average shares as in Table 2, and the results are similar if we use just any one year's data to construct the shares.

[30] We also include the resource tax in the effective price.

A caveat of this microfoundation is that the effective price P_{it} would no longer be a sufficient statistic for all relevant prices if the output share s_{ij} is heterogeneous across individual mines. In the extreme case, each mine may supply coal to only one destination province, as opposed to splitting its output across provinces. Although we do not have mine-level data on output and shipment, we can rule out this extreme case based on news reports where large mines routinely supply to buyers in multiple provinces (Xinhua, 2016; China Securities Journal, 2017; Xinhua, 2018).

Static Supply Assumption

We assume that the coal output decisions are static. According to our interviews with industry practitioners, mine owners plan production based on current prices, and dynamic factors such as the amount of unexcavated coal or interest rates do not appear to be first-order drivers of their production decisions. For context, the coal reserves in China are close to 150 billion tons, while the annual production varies between 3 and 4 billion tons, with a similar amount of new reserves discovered every year.

Supply Function After Policy

Given that we observe only 25 provinces, our model of the policy effects is necessarily parametric. We allow the policy to affect α^S, thereby changing the marginal cost curve once production exceeds a threshold. We estimate the following specification, similar to Ryan (2012):

$$\ln \mathrm{mc}_{it} = \alpha^S \frac{q_{it}}{K_{it}} + \alpha^{S,\text{policy}} \chi_{it}^{\text{policy}} \left(\frac{q_{it}}{K_{it}} - \nu \right)_+ + FE_i^{\text{supply}} + FE_{m(t)}^{\text{supply}} + \xi_{it}, \tag{15}$$

where $\chi_{it}^{\text{policy}}$ is an indicator for the policy period, beginning in April 2016 for Shanxi and May 2016 for other provinces, and $(x)_+ = \max(x, 0)$. The term $\left(\frac{q_{it}}{K_{it}} - \nu \right)_+$ captures the additional effect of the policy on marginal costs when production approaches capacity, with the cutoff ν estimated from the data. This functional form lets the data discipline how sharply marginal costs rise once production exceeds the cutoff, and therefore allows us to quantify the misallocation induced by a command-and-control production restriction. In the extreme case where $\alpha^{S,\text{policy}}$ is large, the policy effectively mimics an efficiently implemented within-province quota, with quota level νK_{it}.[31]

[31] A potential concern with the specification is that the policy in 2016 initially imposed a 276-working-day restriction that was phased out in December (Section 2.3.1), while the 2017 policy did not. Policies in both years increased monitoring and reduced production by limiting overtime production. With limited data, we pool the periods across the two years to estimate the nonlinear effects of the policy. Data suggest that

6.1 Identification

The supply function and the resulting estimation equation can be written in the general form

$$\ln P_{it} = g\left(q_{it}, X_{it}^{\text{supply}}\right) + \xi_{it}, \tag{16}$$

where q_{it} is correlated with ξ_{it}, and there is a vector of exogenous covariates X_{it}^{supply} that accounts for province capacities, locations and time. A sizable literature (e.g., Newey and Powell, 2003) studies the nonparametric identification and estimation of models in this form. In addition to requiring instruments that are mean or quantile independent of the unobservables, the instruments should induce sufficient variations in q.

We use demand shifters in (10) as instruments for supply (Wright, 1928). The excluded instruments include wind generation, the one-month lagged CDD and the exponential of demand residuals, which we assume to be orthogonal to shocks to coal's production costs (MacKay and Miller, 2024; Döpper et al., 2024). We further construct more flexible instruments by replacing each continuous instrument with a set of indicators for whether its value falls within predefined bins.[32] We include these instruments and their interactions with the policy indicator.

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assuming the same policy effect is a reasonable simplification, because the quantity difference due to the 276-day restriction per se is likely small. The total coal production level in 2016 is 3.410 billion tons, and the level in 2017 is 3.491 billion tons, 2% higher. Over the same months, June to November, when the 276-day restriction is active in 2016 but not in 2017, the production levels are also similar, at 1.73 (2016) and 1.77 (2017) billion tons.

[32] For wind, we use bins based on 0.25, 0.5 and 1 TWh (which corresponds with the variable's quantiles of 32.5%, 52.7% and 79.8%). For CDD, we use 0 and 50 degree days (which corresponds with the variable's quantiles of 40.6%, 89.2%). For exponentiated demand residuals, we use 2.5, 5 and 10 thousand tons (which corresponds with the variable's quantiles of 10.4%, 51.9% and 80.2%).

Table 5: Supply Estimation

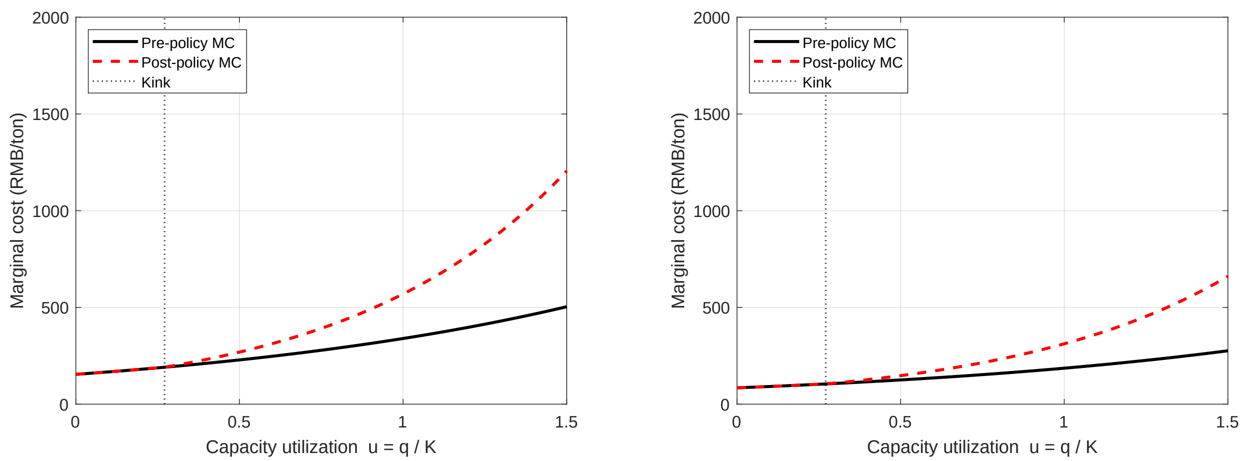
	(1) Baseline	(2) Lower Threshold	(3) Higher Threshold
α^S	1.014***	1.118***	0.705***
	(0.216)	(0.218)	(0.189)
$\alpha^{S,\text{policy}}$	1.007***	0.682***	1.669***
	(0.252)	(0.152)	(0.520)
ν	0.523**	0.290	0.756
	(0.233)		
Province FE	YES	YES	YES
Month FE	YES	YES	YES
N	1200	1200	1200

Notes: The table reports GMM estimates of the common-threshold supply model using fixed coarse indicator-function instruments. Column (1) estimates ν. Columns (2) and (3) re-estimate the slope parameters fixing ν at the column (1) estimate minus and plus one bootstrap standard error, respectively. Standard errors, reported in parentheses, are clustered at the province level. The standard error for ν in column (1) is computed using a province-cluster wild bootstrap. *, **, and *** indicate significance at the 10, 5, and 1 percent levels.

Figure 3: Estimated Aggregate Supply in the Two Largest Coal-Producing Provinces (Shanxi and Inner Mongolia)

(a) Shanxi

(b) Inner Mongolia



The figures compare the aggregate supply functions of Shanxi and Inner Mongolia with and without policy effects in 2017.

6.2 Estimation

Column (1) of Table 5 reports the baseline estimates of (15). The pre-policy slope is $\alpha^S = 1.014$, and the additional policy slope above the threshold is $\alpha^{S,\text{policy}} = 1.007$. The estimated threshold is $\nu = 0.523$, with a bootstrap standard error of 0.233. Columns (2) and (3) assess the robustness of the threshold estimates, re-estimating the remaining supply parameters after fixing ν one standard error below and above the baseline estimate.

To visualize these costs, Figure 3 plots the aggregate marginal cost functions for Shanxi and Inner Mongolia before and after the policy. We interpret these increases as reflecting several factors. First, the policy may impose direct compliance costs, such as halting production for inspections. Second, by preventing low-cost mines from operating above capacity, the policy reallocates output toward higher-cost mines, shifting the production mix toward costlier producers and raising aggregate marginal costs for any given total quantity. Third, when the policy imposes more restrictions on working hours, the mines may have incentive to overwork the labor and capital, increasing the wear and tear.

Finally, we compare the average variable costs with costs reported in financial statements of publicly listed coal mining firms in 2014–2016. We plot these costs in Appendix Figure I.5. The estimated and reported costs are well aligned, with a correlation coefficient of 0.563. For example, we find that Xinjiang and Inner Mongolia have the lowest production costs, consistent with more open-pit mining in these provinces.

7 Counterfactual Simulation

We begin by defining the equilibrium given a set of shocks. Let $\mathcal{J}$ denote the set of coal-consuming provinces, $\mathcal{I}$ for the coal-producing provinces and τ_{ij} for the transportation costs.

Definition. In period t, an equilibrium consists of a set of delivery prices $\{p_{jt}\}_{j\in\mathcal{J}}$ and production quantities $\{Q_{it}\}_{i\in\mathcal{I}}$ that satisfy (10) and (16), where a province j's demand is given by $q_{jt} = \sum_{i\in\mathcal{I}} s_{ij} Q_{it}$, and the price facing producers in province i is $P_{it} = \sum_j s_{ij}\left(p_{jt} - \tau_{ij}\right)$.[33]

In all simulations, we allow foreign imports to respond endogenously, following an estimated import supply function.[34]

The results in this section are based on the baseline demand estimate in column (5) of Table 4 and supply estimate in column (1) of Table 5. In Appendix J, we report the results

[33] We can establish the equilibrium existence by first constructing closed intervals of prices and quantities on which the equilibrium conditions map the extrema of the intervals back into the intervals using the monotonicity of the demand and supply functions and the fact that their range is $(0, \infty)$. Then an application of Brouwer's fixed point theorem shows existence. For uniqueness, we do not find multiple equilibria in our simulations.

[34] Our shipment data show that the share of imports to each province is also stable over time and allows us to construct a similar effective price for imports. We estimate the import supply elasticity in a log-log specification. An OLS regression based on data from 2014 to 2017 (48 months) yields an estimate of 1.35, with a robust standard error of 0.15. A corresponding IV estimation produces nearly identical results.

based on the alternative specifications of supply in columns (2) and (3) of Table 5, which are similar to the baseline results here.

7.1 Economic Surplus and Pollution Damages in the Welfare Function

The supply reduction policy of the reform has two main effects: (1) it reduces capacity by closing small mines and halting new entry, and (2) it restricts production by closely monitoring remaining mines to prevent production above approved capacity. Because we lack mine-level data on entry and exit, we take capacity as given and focus on the effects of production restriction. Our estimates of economic surplus should be interpreted as the sum of consumer surplus and variable profits, which do not account for investment or fixed costs.

We also compute the pollution damages of coal. We estimate how changes in coal consumption affect the air quality (PM2.5 and PM10 concentrations) in a province, and we translate the quality difference into local health cost changes and aggregate across provinces. We also include environmental costs of coal production based on water depletion. The details are in Appendix H.

Our approach may understate the benefits from reducing coal production because it does not quantify all environmental damages associated with coal. These omitted damages include pollution from coal production (Chu et al., 2023), transportation, and storage (Jha and Muller, 2018), as well as air pollution from excavation and land degradation.

We nevertheless believe that our estimates capture the first-order effects of pollution damages. First, reducing coal consumption and the resulting air pollution likely generate the largest welfare gains, given the large affected population, and were plausibly a central objective of the supply reduction policy (Feng et al., 2019). Second, our estimate of pollution damages during coal production, RMB 86.6 per ton of coal, is larger than the damage value implied by the resource tax. The resource tax is designed primarily to address localized pollution and ecological damage from coal production, and its rate is capped at 10% of coal sales revenue. With an average coal price of about RMB 532 per ton in our sample, this cap implies a maximum damage value of RMB 53.2 per ton, which is below our estimate.

We also do not assign an explicit monetary value to carbon emissions in the baseline welfare function. This choice is motivated by both measurement and institutional considerations. As noted in the Introduction, estimates of the social cost of carbon are highly sensitive to assumptions about discount rates, climate damages, and tipping points, making any single value difficult to justify in this setting. More importantly, climate change is not one of the objectives motivating the 2016 production restriction. We therefore interpret the welfare function in this section as a policy-relevant objective for China's coal-market intervention, rather than as a complete global social welfare function. This does not mean that carbon damages are unimportant. Rather, we avoid imposing a particular social cost of carbon in the baseline analysis and instead address these unmeasured damages in Section 7.3.2, where we allow for a general increasing damage function $L(Q)$ that can capture carbon emissions and other unmeasured social costs associated with aggregate coal use.

7.2 Production Restriction

To simulate the no-policy effects, we remove the estimated policy effect in supply and recompute the expected equilibrium outcomes by setting $\alpha^{S,\text{policy}} = 0$. We assume that the demand and supply shocks at the province level follow an AR(1) process, and we also simulate forward the processes of CDD and wind generation.[35] We report the 2.5%–97.5% range and the average of market outcomes based on 300 simulation paths.

Figure 4 compares the average equilibrium prices and production under the no-policy simulation with observed outcomes under the policy, holding fixed the capacity as observed in the data. The simulations align closely with the data before the policy. We further show in Appendix I.5 that the model can match province-level production despite the parsimonious specification. Absent the policy, prices would have been substantially lower and production substantially higher.

Welfare Effects

Table 6 presents the changes in welfare in a no-policy scenario, relative to the observed policy. We focus on 2017 because inventory adjustment was unusually large in 2016: between April and December 2016, a substantial inventory drawdown contributed to the consumption by coal's end users. By 2017, inventory levels had largely stabilized, so consumption more closely tracked production. This makes 2017 the appropriate period to apply our equilibrium framework, in which demand reflects consumption and supply maps production to consumption without a sizable inventory-change component. We simulate the market outcomes with and without the policy effects using the estimated time-series processes for shocks and other covariates conditional on their values in the last month of 2016.[36]

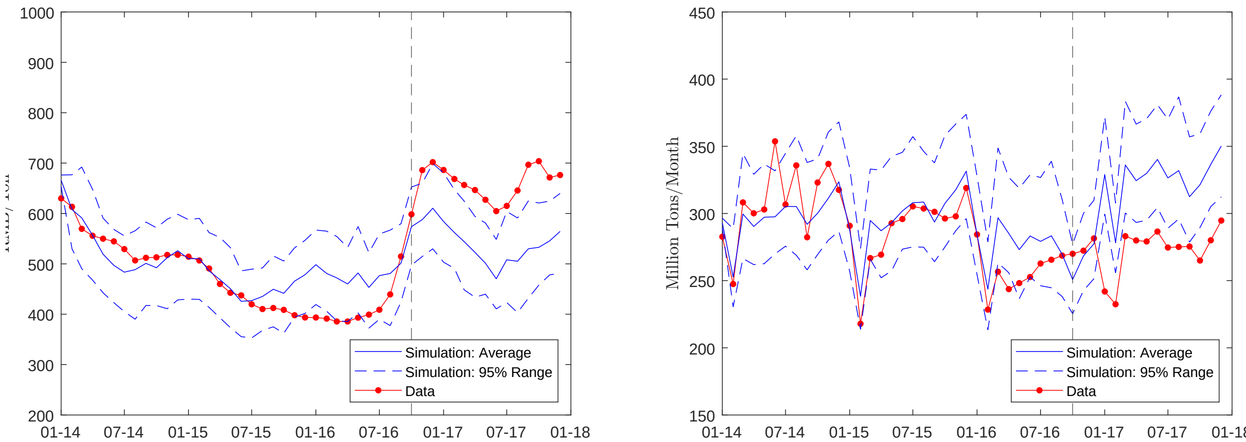
Table 6 shows that removing the policy lowers delivery prices by RMB 145/ton and raises domestic production by 668 million tons in 2017. Consumers gain RMB 560 billion from lower coal prices, while producers lose RMB 288 billion and resource-tax revenue falls by RMB 14 billion. The no-policy scenario raises economic surplus by RMB 259 billion, but pollution

[35] Appendix F reports the estimates of these processes.

[36] It is possible that the time-series processes before and after the policy may differ. We find that all of our results are qualitatively robust to using the processes estimated based on 2017 data alone.

Figure 4: Prices and Production With and Without the Policy

(a) Prices (b) Production



The figures plot the simulated prices and production of Shanxi and Inner Mongolia from 300 simulations in a no-policy counterfactual. We plot the simulated 2.5%–97.5% range, the average and the data paths (with policy). The vertical lines indicate the start of the policy.

damages increase by RMB 309 billion because of higher coal production and consumption. Measured welfare is therefore RMB 50 billion lower without the policy. Appendix Figure I.7 breaks down the surplus changes at the province level, consistent with the policy benefiting coal-producing provinces and hurting coal-consuming provinces. The welfare gain from the policy would be larger if we take into account additional forms of pollution damages or the social cost of carbon.

Table 6: No-policy Effects

Δ Delivery price (RMB/ton)	-145
	[-157, -132]
Δ Domestic coal production (million tons/year)	668
	[589, 762]
Δ Coal consumption (million tons/year)	621
	[545, 712]
Δ Economic Surplus and Pollution Damages (RMB bn.)	
Economic Surplus	259
	[222, 296]
Consumer	560
	[489, 629]
Producer	-288
	[-320, -255]
Resource Tax	-14
	[-16, -12]
Avoided Pollution Damages	-309
	[-341, -282]
Δ Welfare	-50
	[-73, -29]

Notes: The table reports changes relative to the observed policy in 2017. Square brackets report the 2.5–97.5 percent range across simulations. Avoided pollution damages include local air-pollution health benefits and water depletion, and exclude carbon.

Allocative Inefficiency

We next quantify the policy's inefficiency relative to an efficiently implemented quota. We note two types of inefficiencies. First, our estimates suggest that the policy raises inframarginal costs and leads to within-province allocative inefficiency. Second, our estimates indicate greater production restrictions in larger, lower-cost provinces such as Inner Mongolia, indicating cross-province inefficiency. Although cap-and-trade, or a productivity-based production restriction, is likely infeasible given the institutional constraints, understanding these inefficiencies not only sheds light on the welfare costs of the policy, but also enables us to apply the ranking criterion in Section 3 to compare the policy with a hypothetical tax.

We first decompose the inefficiencies caused by the supply function shift in each province. Holding the 2017 equilibrium quantity fixed in each simulation path, we re-compute producer surplus using the pre-policy supply curve integrated up to that quantity. The within-province inefficiency is therefore the increase in producer surplus under the pre-policy supply curve relative to the observed policy, holding fixed the same provincial quantities. This within-province distortion reduces producer surplus by RMB 136 billion.

Next, we measure the total inefficiency. An efficient quota allocation across provinces requires the marginal surplus loss from an additional unit of reduction to be equalized across provinces. Therefore, holding fixed a time series of simulated shocks in 2017, for any total reduction ΔQ, the efficient allocation is the one implied by a uniform tax that achieves the same total reduction. We use the following procedure to find the efficient allocation: in each simulation based on the post-policy supply function, we (i) solve for the tax that delivers the same aggregate reduction in equilibrium, (ii) fix the quantities produced in each province under this tax, and (iii) compute economic surplus by integrating demand and the pre-policy supply functions up to those efficient quantities. The total producer-surplus gain from correcting within- and cross-province misallocation is RMB 212 billion. Therefore, 64% of the misallocation is within province.

Table 7: Tax Effects (RMB bn.) Relative to the Observed Policy

	(1) No Policy	(2) Optimal	(3) Optimal, Weighted
Δ Consumer Surplus	560	-602	-368
Δ Producer Surplus	-288	-660	-615
Δ Tax Revenue	-14	1270	1071
Δ Saved Pollution Damages	-309	253	164
ΔW	-50	261	251

Notes: The table reports changes in 2017 relative to the observed policy. Saved pollution damages include local air-pollution health benefits and water depletion, and exclude carbon. The tax levels in columns (1), (2), and (3) are 0, 417, and 331 RMB/ton, respectively.

7.3 Is Tax More Efficient?

In Section 2.3.2, we discussed the political reasons why production restrictions, as opposed to a tax, are used to implement the reform. Yet, the economic question remains: is it more efficient to use a tax? Section 3 shows that whether the best tax generates a higher welfare than the best production restriction relies on the slopes of the demand, cost and marginal damages as well as their variances. We present two sets of results that point to the better efficiency of tax.

7.3.1 Optimal Tax Maximizing Economic Surplus and Avoided Pollution Damages

We first consider a national government's objective function that takes into account the impact on economic surplus and pollution damages that we can measure:

$$W = E(\text{consumer surplus} + \text{producer surplus} + \text{tax revenue} - \text{pollution damages}), \quad (17)$$

where the damages correspond with the monetary values of the environmental damages in Section 7.2. An optimal counterfactual tax is set at the beginning of 2017 and maximizes the objective above, where the expected value of each component in the function is based on the estimated AR(1) processes of demand and marginal cost shocks and the realized values in the last month of 2016.

Column (2) of Table 7 reports the welfare effects of the optimal tax. The optimal tax is RMB 417/ton and raises W by RMB 261 billion relative to the observed policy. Under this tax, consumer surplus falls by RMB 602 billion and producer surplus falls by RMB 660 billion, but these losses are more than offset by RMB 1,270 billion in tax revenue and RMB 253 billion in additional saved pollution damages. Absent the hurdles to redistribute the tax revenues (Section 2), rebating the tax revenue would offset most of the decline in economic surplus.

This gain arises because the observed equilibrium is on relatively inelastic portions of both demand and supply, allowing taxation to generate substantial revenue before higher tax rates induce more elastic responses from either side of the market.

Anticipation of the Demand and Supply Shocks An assumption in the calculation above is that the unobservables in the demand and supply functions are also unknown to the market participants or policymakers. This assumption is important for correctly defining expected welfare, but our qualitative results on the efficiency of the tax are likely robust to stronger information assumptions. In the extreme case where the unobservables are fully known to buyers, sellers and the government, a tax would generate less allocative inefficiency and would be unambiguously more efficient than a production restriction for the same quantity target.

7.3.2 Optimal Tax with a General Damage Function

Next, we significantly strengthen the efficiency comparison using the stochastic dominance approach from Section 3.5. A more general government objective may account for broader effects of coal consumption, such as carbon emissions, through an increasing damage function $L(Q)$ on total output. There may also be other pollution externalities that we cannot model directly but that can be approximated as an increasing function of aggregate production Q.[37]

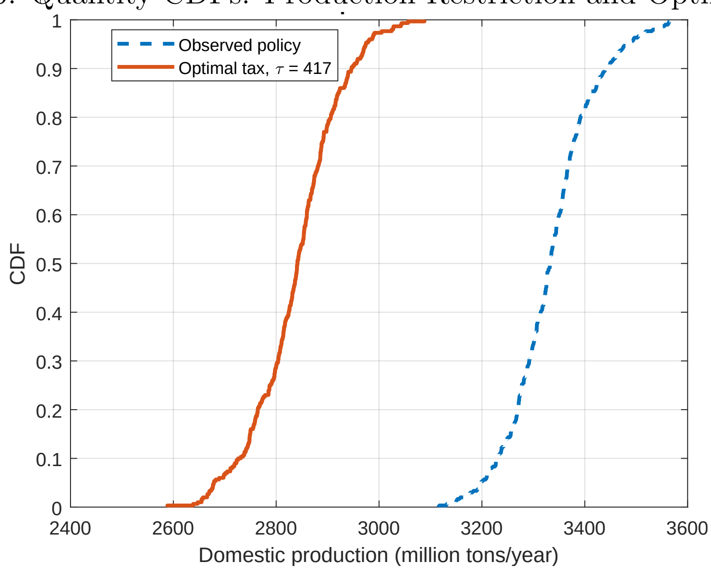
[37] Leakage is an externality that we do not explicitly account for. Next to coal (65%), hydro (20%) had the second largest share in the power generation mix in 2016, followed by wind (4%) and nuclear (4%). Natural gas accounted for 3% of the generation mix (Göß and Niggemeier, 2017). In the long run, reduced coal consumption, especially in electricity generation, is likely to be offset by both the growth of renewable generations and increased natural gas use (EIA, 2022). Generations from these sources typically have lower local pollution intensities and carbon emissions than coal. Therefore, such substitution would still be expected to reduce the overall pollution damage when coal consumption falls.

The augmented welfare is

$$\begin{aligned}W^\star =&E(\text{consumer surplus} + \text{producer surplus} + \text{tax revenue} - \text{pollution damages} - L(Q))\\ =&W - E(L(Q)).\end{aligned}$$

We apply the procedure in Section 3.5 to compare $W^\star$ under the observed policy with that of the optimal tax from the previous section: the tax achieves a higher W than the observed policy, and we check whether the quantity distribution under the tax is stochastically dominated. Figure 5 confirms this: the output CDF under the optimal tax lies everywhere above the CDF under the production restriction.[38] Since $W_{\text{tax}} > W_{\text{restriction}}$ and $E(L(Q_{\text{tax}})) \leq E(L(Q_{\text{restriction}}))$ for any increasing L, the tax achieves a higher $W^\star$. Moreover, a range of taxes around the optimal tax also achieve higher W with stochastically dominated quantity distributions.

Figure 5: Quantity CDFs: Production Restriction and Optimal Tax



We plot the CDFs of 2017 production under the observed policy and the optimal tax.

[38]There are several ways to formally test the stochastic dominance. First, for a simple but formal statistical test, we test an even stronger condition, which is that the quantity under the optimal tax based on the estimated parameters is statistically smaller than under the policy for each simulated path of shocks. More generally, given sufficiently many simulation draws, we can ignore the simulation errors and compute the maximal difference between the two CDFs over their support. The 95% confidence interval of this difference can be simulated using the asymptotic distribution of the estimated parameters.

7.4 Policy Preferences

The results above show that the government was willing to forgo at least RMB 261 billion in welfare gains when choosing the production restriction over the optimal tax. As Section 2 and Section 3 show, policy priorities of improving producer surplus and reducing pollution damages may explain this choice.

To compute optimal taxes when the government has policy preferences for some components of the welfare function, we focus on the following weighted welfare function based on W without the unknown damage term L:

$$\tilde{W}(w_{\text{PS}}, w_{\text{damage}}) = E(\text{consumer surplus} + w_{\text{PS}} \cdot \text{producer surplus} \\ + \text{tax revenue} - w_{\text{damage}} \cdot \text{pollution damages}).$$

In the above, a higher w_{PS} makes the production restriction policy more appealing, whereas the effect of a higher w_{damage} is ambiguous. We thus consider a range of weight pairs, $(w_{\text{PS}}, w_{\text{damage}})$. For each pair, we solve for the optimal tax and then evaluate whether welfare is higher under that tax than under the observed production restriction policy.

We find that, for a given value of w_{damage}, the production restriction generates a higher $\tilde{W}$ than the optimal tax when w_{PS} exceeds a threshold. Table 8 reports these thresholds. When the policy maker values consumer surplus, tax revenue and pollution damages equally ($w_{\text{damage}} = 1$), valuing producer surplus 40.8% higher justifies the observed production restriction against even the optimal tax.

This result reflects several important realities in China's coal industry.[39] First, the profits of coal mining firms, many of which are state-owned, are not entirely private rents. Part of the state owned firms' after-tax profits flows back to the government. This gives the government a direct fiscal stake in coal-firm profitability. Second, the surplus helps to stabilize employment and avoid worker resettlement. The falling prices in 2014-2015 forced provincial governments to reduce local taxes and expand safety net (Shanxi, 2015, 2016). Third, the profits reduce the need for external financing when the government encourages investment in efficient upgrades (Yuan, 2025). The government chooses the production restriction policy despite the significant increase in tax revenues under the more optimal tax, which reflects the significant hurdles for redistributing the tax revenues (Section 2).

[39] This result is different from some policy preference estimates in other settings. For example, Lim and Yurukoglu (2018) estimates that government places much higher weight on consumer surplus than on utilities in the utility rate-setting context in the US.

Table 8: Policy Weights That Rationalize Production Restrictions Over Taxes

w_{damage}	1.00	1.25	1.50	1.75	2.00
$w_{\text{PS}} \geq$	1.408	1.491	1.601	1.732	1.878

Notes: The best-tax levels corresponding to the columns are 325, 425, 525, 625, 650 RMB/ton.

We also note that placing greater weight on pollution damages raises the threshold value of w_{PS} needed to rationalize the observed policy relative to a tax. The intuition is related to our finding in Section 7.3.2: given our demand and supply estimates, a tax can often achieve higher economic surplus while reducing quantities, and therefore pollution damages, more than the production restriction policy. As a result, a tax becomes more attractive at higher values of w_{damage}, so a larger weight on producer surplus is required for the government to prefer production restrictions over a tax.

This result also has an important implication for our interpretation of the lower bound. As discussed above, our estimates likely understate total pollution damages. If the omitted damages are proportional to the damages we measure and enter the government objective function in the same way, then the true value of w_{damage} would be higher. In that case, the production restriction would be even harder to rationalize relative to a tax, implying that the lower bound on w_{PS} would exceed 1.408.

8 Conclusion

We study China's 2016 coal industry reform, which restricted coal production through mine closures and output controls. Empirically, we find that the policy reduced coal output, raised prices, and shifted surplus from consumers to producers. Using an estimated equilibrium model of provincial coal demand and supply, we quantify these distributional effects and compare the observed production restriction with a tax. We also develop an approach to compare the two instruments when some of the pollution damages are not measured but increase in the total quantity.

Our welfare analysis shows that the production restriction was not the surplus-maximizing way to reduce coal consumption under equal welfare weights. A tax could have reduced aggregate output while raising total welfare, even after accounting for environmental and health benefits from avoided coal use. This finding highlights the efficiency cost of using production restrictions rather than price-based instruments. At the same time, implementing such a tax would have required substantial fiscal and institutional reform, whereas production restrictions could be imposed more easily.

We further show that the observed policy can also be rationalized by policy preferences.

The reform documents emphasize not only pollution reduction, but also coal-industry profitability and economic stability in producing regions. Consistent with these objectives, we show that even an optimal tax no longer dominates the observed policy when the government places sufficiently high weight on producer surplus. More broadly, we provide an empirical framework to bound these weights.

The results also underscore the challenges of transitioning to a low-carbon economy given the institutional constraints. Consistent with this observation, China has made massive investments in renewable energy. In 2024, for the first time, the share of coal in electricity generation fell below 60%, primarily driven by the growth of low-cost electricity production from renewable sources, as opposed to reduced coal consumption (Howe, 2024).

References

Adão, Rodrigo, John Sturm Becko, Arnaud Costinot, and Dave Donaldson. 2026. "Why Is Trade Not Free? A Revealed Preference Approach." Manuscript, revision resubmitted to Econometrica.

Adar, Zvi, and James M. Griffin. 1976. "Uncertainty and the Choice of Pollution Control Instruments." *Journal of Environmental Economics and Management*, 3(3): 178–188.

Aden, Nathaniel, David Fridley, and Nina Zheng. 2009. "China's Coal: Demand, Constraints, and Externalities." Ernest Orlando Lawrence Berkeley National Laboratory, Berkeley, CA.

Aghion, Philippe, Jing Cai, Mathias Dewatripont, Luosha Du, Ann Harrison, and Patrick Legros. 2015. "Industrial policy and competition." *American Economic Journal: Macroeconomics*, 7(4): 1–32.

Alder, Simon, and Illenin Kondo. 2019. "Political Distortions and Infrastructure Networks in China: A Quantitative Spatial Equilibrium Analysis." Manuscript.

Alder, Simon, Lin Shao, and Fabrizio Zilibotti. 2016. "Economic reforms and industrial policy in a panel of Chinese cities." *Journal of Economic Growth*, 21: 305–349.

Aldy, Joseph E., Todd D. Gerarden, and Richard L. Sweeney. 2023. "Investment Versus Output Subsidies: Implications of Alternative Incentives for Wind Energy." *Journal of the Association of Environmental and Resource Economists*, 10(4): 981–1016.

Almond, Douglas, and Shuang Zhang. 2021. "Carbon-Trading Pilot Programs in China and Local Air Quality." *AEA Papers and Proceedings*, 111: 391–395.

Almond, Douglas, Yuyu Chen, Michael Greenstone, and Hongbin Li. 2009. "Winter heating or clean air? Unintended impacts of China's Huai river policy." *American Economic Review*, 99(2): 184–190.

American Geosciences Institute. n.d.. "What Are the Different Types of Coal?" `https://profession.americangeosciences.org/society/intersections/faq/what-are-the-different-types-of-coal/`.

Bai, Jie, Panle Jia Barwick, Shengmao Cao, and Shanjun Li. 2020. "Quid pro quo, knowledge spillover, and industrial quality upgrading: Evidence from the Chinese auto industry." National Bureau of Economic Research.

Bargain, Olivier, Mathias Dolls, Dirk Neumann, Andreas Peichl, and Sebastian Siegloch. 2014. "Tax-Benefit Revealed Social Preferences in Europe and the US." *Annals of Economics and Statistics*, (113/114): 257–289.

Barwick, Panle Jia, Myrto Kalouptsidi, and Nahim Bin Zahur. 2019. "China's industrial policy: An empirical evaluation." National Bureau of Economic Research.

Barwick, Panle Jia, Shanjun Li, Deyu Rao, and Nahim Bin Zahur. 2021. "The Healthcare Cost of Air Pollution: Evidence from the World's Largest Payment Network." National Bureau of Economic Research Working Paper 24688 `http://www.nber.org/papers/w24688`.

Baumol, William J. 1972. "On Taxation and the Control of Externalities." *American Economic Review*, 62(3): 307–322.

Bell, Ruth G, and Clifford Russell. 2002. "Environmental policy for developing countries." *Issues in Science and Technology*, 18(3): 63–70.

Berman, Eli, and Linda TM Bui. 2001. "Environmental regulation and productivity: evidence from oil refineries." *Review of Economics and Statistics*, 83(3): 498–510.

Bie, Fan. 2015. "Initial Long-Term Coal-Power Contract Price Set at RMB 520 per Ton." *China Energy News*, `https://paper.people.com.cn/zgnyb/html/2015-02/02/content_1530276.htm`.

Blackman, Allen, and Winston Harrington. 2000. "The use of economic incentives in developing countries: Lessons from international experience with industrial air pollution." *Journal of Environment & Development*, 9(1): 5–44.

Blackman, Allen, Zhengyan Li, and Antung A Liu. 2018. "Efficacy of command-and-control and market-based environmental regulation in developing countries." *Annual Review of Resource Economics*, 10(1): 381–404.

Bloomberg. 2022. "China's Coal Expansion." *Bloomberg Professional Services* https://www.bloomberg.com/professional/insights/trading/chinas-coal-expansion/.

BOCI. 2017. "BOCI Global Insight." Bank of China International, Hong Kong https://www.bocigroup.com/UploadFiles/file/20170928/636422113243415831432157 5.pdf.

Bourguignon, François, and Amedeo Spadaro. 2012. "Tax–Benefit Revealed Social Preferences." *Journal of Economic Inequality*, 10(1): 75–108.

Brandt, Loren, John Litwack, Elitza Mileva, Luhang Wang, Yifan Zhang, and Luan Zhao. 2020. *China's productivity slowdown and future growth potential.* The World Bank.

Burke, Paul J, and Hua Liao. 2015. "Is the price elasticity of demand for coal in China increasing?" *China Economic Review*, 36: 309–322.

Cao, Yanou, and Xiujuan Feng. 2017. "Study on Coal Regional Competitiveness Part I: Research on Coal Transportation Channels and Freight in China." *China Chengxin International*, https://www.chinaratings.com.cn/CreditResearch/Industry/TopicReport/117924.html.

CCTV News. 2024. "Stable Foundations for Ensuring Power Supply in the Summer Peak Season." https://news.cctv.com/2024/08/02/ARTIf7tJOZLVcLSQtHObmnnu240802.shtml.

Chen, Qiaoyi, Zhao Chen, Zhikuo Liu, Juan Carlos Suárez Serrato, and Daniel Yi Xu. 2025. "Regulating conglomerates: Evidence from an energy conservation program in china." *American Economic Review*, 115(2): 408–447.

Chen, Yuyu, Avraham Ebenstein, Michael Greenstone, and Hongbin Li. 2013. "Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy." *Proceedings of the National Academy of Sciences*, 110(32): 12936–12941.

China National Coal Association. 2018. "2017 Coal Industry Development Annual Report." China National Coal Association https://www.cctd.com.cn/uploadfile/2018/0328/20180328093649264.pdf.

China Securities Journal. 2017. "NDRC to Introduce Policies to Accelerate Coal-Power Integration." https://www.nea.gov.cn/2017-11/27/c_136782628.htm.

Chu, Yin, J Scott Holladay, Yun Qiu, Xian-Liang Tian, and Maigeng Zhou. 2023. "Air pollution and mortality impacts of coal mining: Evidence from coalmine accidents in China." *Journal of Environmental Economics and Management*, 121: 102846.

Clark, Alex, and Weirong Zhang. 2022. "Estimating the employment and fiscal consequences of thermal coal phase-out in China." *Energies*, 15(3): 800.

Criscuolo, Chiara, Ralf Martin, Henry G Overman, and John Van Reenen. 2019. "Some causal effects of an industrial policy." *American Economic Review*, 109(1): 48–85.

Crocker, Thomas D. 1966. "The structuring of atmospheric pollution control systems." In *The Economics of Air Pollution.* , ed. Harold Wolozin, 61–86. New York:W.W. Norton & Company.

Dales, John H. 1968. *Pollution, Property and Prices.* Toronto:University of Toronto Press.

Delabastita, Vincent, and Michael Rubens. 2022. "Monopsony and Industrialization." Working paper.

Demirer, Mert, and Michael Rubens. 2025. "Welfare Effects of Buyer and Seller Power." National Bureau of Economic Research.

Dickie, Mark, and Gregory A. Trandel. 1996. "Comparing Specific and Ad Valorem Pigouvian Taxes and Output Quotas." *Southern Economic Journal*, 63(2): 388–405.

Döpper, Hendrik, Alexander MacKay, Nathan Miller, and Joel Stiebale. 2024. "Rising Markups and the Role of Consumer Preferences." *Journal of Political Economy*. Forthcoming.

Ebenstein, Avraham, Maoyong Fan, Michael Greenstone, Guojun He, and Maigeng Zhou. 2017. "New Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy." *Proceedings of the National Academy of Sciences*, 114(39): 10384–10389.

EIA. 2022. "China increased electricity generation annually from 2000 to 2020." *Today in Energy* `https://www.eia.gov/todayinenergy/detail.php?id=53959`.

EIA. 2023. "The Most Coal Is Stockpiled at U.S. Power Plants Since 2020." *Today in Energy*, `https://www.eia.gov/todayinenergy/detail.php?id=62684`.

Eisenberg, Tom. 2019. "Regulatory Distortions and Capacity Investment: The Case of China's Coal Power Industry." Working paper.

Eisenberg, Tom. 2024. "Missing data and the effects of market deregulation: Evidence from Chinese coal power." *International Journal of Industrial Organization*, 93: 103054.

Fajgelbaum, Pablo, Cecile Gaubert, Nicole Gorton, Eduardo Morales, and Edouard Schaal. 2024. "Political Preferences and Transport Infrastructure: Evidence from California's High-Speed Rail." National Bureau of Economic Research NBER Working Paper 31438. Revised July 2024.

Fan, Jingting, and Ben Zou. 2021. "Industrialization from scratch: The "Construction of Third Front" and local economic development in China's hinterland." *Journal of Development Economics*, 152: 102698.

Fell, Harrison, Ian A MacKenzie, and William A Pizer. 2012. "Prices versus quantities versus bankable quantities." *Resource and Energy Economics*, 34(4): 607–623.

Feng, Yueyi, Miao Ning, Yu Lei, Yamei Sun, Wei Liu, and Jinnan Wang. 2019. "Defending blue sky in China: Effectiveness of the "Air Pollution Prevention and Control Action Plan" on air quality improvements from 2013 to 2017." *Journal of Environmental Management*, 252: 109603.

Fowlie, Meredith, Mar Reguant, and Stephen P Ryan. 2016. "Market-based emissions regulation and industry dynamics." *Journal of Political Economy*, 124(1): 249–302.

Francois, Joseph, and Douglas R. Nelson. 2014. "Political Support for Trade Policy in the European Union." *European Journal of Political Economy*, 36: 243–253.

Fu, Shihe, V Brian Viard, and Peng Zhang. 2022. "Trans-boundary air pollution spillovers: Physical transport and economic costs by distance." *Journal of Development Economics*, 155: 102808.

Gao, Ge. 2017. "NDRC: Coal-Fired Power Companies That Fail to Sign or Fulfill Medium- and Long-Term Contracts Will Be Penalized." *Yicai Global*, `https://www.yicai.com/news/5264060.html`.

Gao, Hang, and Johannes Van Biesebroeck. 2014. "Effects of Deregulation and Vertical Unbundling on the Performance of China's Electricity Generation Sector." *Journal of Industrial Economics*, 62(1): 41–76.

Gawande, Kishore, and Usree Bandyopadhyay. 2000. "Is Protection for Sale? Evidence on the Grossman–Helpman Theory of Endogenous Protection." *Review of Economics and Statistics*, 82(1): 139–152.

Gawande, Kishore, Pravin Krishna, and Marcelo Olarreaga. 2009. "What Governments Maximize and Why: The View from Trade." *International Organization*, 63(3): 491–532.

Giorcelli, Michela. 2019. "The long-term effects of management and technology transfers." *American Economic Review*, 109(1): 121–52.

Giorcelli, Michela, and Bo Li. 2021. "Technology transfer and early industrial development: Evidence from the Sino-Soviet alliance." National Bureau of Economic Research.

Goldberg, Pinelopi Koujianou, and Giovanni Maggi. 1999. "Protection for Sale: An Empirical Investigation." *American Economic Review*, 89(5): 1135–1155.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. "Bartik Instruments: What, When, Why, and How." *American Economic Review*, 110(8): 2586–2624 `https://doi.org/10.1257/aer.20181047`.

Göß, Simon, and Marie-Louise Niggemeier. 2017. "Energy Transition in the Power Sector in China: State of Affairs in 2016: Review on the Developments in 2016 and an Outlook." Agora Energiewende and China National Renewable Energy Centre Analysis 116/06-A-2017/EN, Berlin, Germany and Beijing, China. Revised version. Prepared by Energy Brainpool on behalf of Agora Energiewende and China National Renewable Energy Centre `https://www.agora-energiewende.de/`.

Goulder, Lawrence H., Xianling Long, Chenfei Qu, and Da Zhang. forthcoming. "China's Nationwide CO2 Emissions Trading System: A General Equilibrium Assessment." *American Economic Journal: Economic Policy*.

Goulder, Lawrence H., Xianling Long, Jieyi Lu, and Richard D. Morgenstern. 2022. "China's Unconventional Nationwide CO2 Emissions Trading System: Cost-Effectiveness and Distributional Impacts." *Journal of Environmental Economics and Management*, 111: 102561.

Gowrisankaran, Gautam, Charles He, Eric A Lutz, and Jefferey L Burgess. 2015. "Productivity, safety, and regulation in underground coal mining: Evidence from disasters and fatalities." National Bureau of Economic Research.

Greenstone, Michael. 2002. "The impacts of environmental regulations on industrial activity: Evidence from the 1970 and 1977 clean air act amendments and the census of manufactures." *Journal of Political Economy*, 110(6): 1175–1219.

Greenstone, Michael, John A List, and Chad Syverson. 2012. "The effects of environmental regulation on the competitiveness of US manufacturing." National Bureau of Economic Research.

Gu, Alun, and Sun Li. 2017. "Actual influence cost estimation of water resources in coal mining and utilization in China." *Energy Procedia*, 142: 2454–2460.

Guo, Xiaodan, and Junji Xiao. 2022. "Welfare analysis of the subsidies in the Chinese electric vehicle industry." *Journal of Industrial Economics*. Forthcoming.

Hadar, Josef, and William R. Russell. 1969. "Rules for Ordering Uncertain Prospects." *American Economic Review*, 59(1): 25–34.

Hanlon, W Walker. 2020. "The Persistent Effect of Temporary Input Cost Advantages in Shipbuilding, 1850 to 1911." *Journal of the European Economic Association*, 18(6): 3173–3209.

He, Guojun, Shaoda Wang, and Bing Zhang. 2020. "Watering down environmental regulation in China." *Quarterly Journal of Economics*, 135(4): 2135–2185.

Helfand, Gloria E. 1991. "Standards versus Standards: The Effects of Different Pollution Restrictions." *American Economic Review*, 81(3): 622–634.

Hendren, Nathaniel. 2020. "Measuring Economic Efficiency Using Inverse-Optimum Weights." *Journal of Public Economics*, 187: 104198.

Hepburn, Cameron. 2006. "Regulation by prices, quantities, or both: a review of instrument choice." *Oxford Review of Economic Policy*, 22(2): 226–247.

Hervé-Mignucci, Morgan, Xueying Wang, David Nelson, and Uday Varadarajan. 2015. "Slowing the growth of coal power in China: The role of finance in state-owned enterprises." Climate Policy Initiative.

Howe, Colleen. 2024. "China is building two-thirds of new wind and solar globally, report says." *Reuters*. Updated July 11, 2024 `https://www.reuters.com/sustainability/climate-energy/china-coal-generation-share-record-low-may-renewables-hit-new-highs-analysis-2024-07-`

Hu, Jun. 2014. "2014 Coal Contract Volume Reaches 1.9 Billion Tons." *China Energy News*, `https://paper.people.com.cn/zgnyb/html/2014-01/20/content_1382438.htm`.

Hu, Yongqi. 2017. "China allots environment tax revenue to local governments." *China Daily (Chinese edition)*, https://cn.chinadaily.com.cn/2017-12/27/content_35394181.htm.

ICBC. 2016. "Shenhua Coal Adjusts Pricing Model Again in August: First Disclosure of Pricing Mechanism." https://www.icbc-ltd.com/icbc/.

International Energy Network. 2016*a*. "Energy Market Insights: Analysis of Recent Developments in Coal Pricing Strategies." https://www.in-en.com/article/html/energy-2257990.shtml.

International Energy Network. 2016*b*. "Method for Calculating Long-Term Coal Contract Prices (with Formula)." *Sohu*, https://www.sohu.com/a/119052248_505851.

Ito, Koichiro, and Shuang Zhang. 2020. "Willingness to Pay for Clean Air: Evidence from Air Purifier Markets in China." *Journal of Political Economy*, 128(5): 1627–1672.

Jaffe, Adam B, Steven R Peterson, Paul R Portney, and Robert N Stavins. 1995. "Environmental regulation and the competitiveness of US manufacturing: what does the evidence tell us?" *Journal of Economic Literature*, 33(1): 132–163.

Jha, Akshaya. 2023. "Dynamic Regulatory Distortions: Coal Procurement at US Power Plants." Working paper.

Jha, Akshaya, and Nicholas Z Muller. 2018. "The local air pollution cost of coal storage and handling: Evidence from US power plants." *Journal of Environmental Economics and Management*, 92: 360–396.

Juhász, Réka. 2018. "Temporary protection and technology adoption: Evidence from the napoleonic blockade." *American Economic Review*, 108(11): 3339–3376.

Kalouptsidi, Myrto. 2018. "Detection and impact of industrial subsidies: The case of Chinese shipbuilding." *Review of Economic Studies*, 85(2): 1111–1158.

Kline, Patrick, and Enrico Moretti. 2014. "Local economic development, agglomeration economies, and the big push: 100 years of evidence from the Tennessee Valley Authority." *Quarterly Journal of Economics*, 129(1): 275–331.

Koenig, Evan F. 1985. "Indirect Methods for Regulating Externalities under Uncertainty." *Quarterly Journal of Economics*, 100(2): 479–493.

Lai, Shasha. 2016. "Coal Industry's 276-Workday Production Restriction Policy Begins to Loosen." *Caixin*, https://companies.caixin.com/2016-11-11/101006548.html.

Lane, Nathaniel. 2022. "Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea." *Quarterly Journal of Economics*, 137(4): 2401–2465.

Lashkaripour, Ahmad, and Vova Lugovskyy. 2018. "Scale economies and the structure of trade and industrial policy." Mimeo, Indiana University.

Li, Chunlian. 2014. "Domestic Coal and Imported Coal Prices Begin to Invert: Imported Coal Transactions Have Fallen into Difficulty." *Securities Daily*, https://energy.people.com.cn/n/2014/0307/c71661-24556476.html.

Lim, Claire S. H., and Ali Yurukoglu. 2018. "Dynamic Natural Monopoly Regulation: Time Inconsistency, Moral Hazard, and Political Environments." *Journal of Political Economy*, 126(1): 263–312.

Liu, Ming-Hua, Dimitris Margaritis, and Yang Zhang. 2013. "Market-driven coal prices and state-administered electricity prices in China." *Energy Economics*, 40: 167–175.

Li, Xiaoyu. 2016. "CPPCC Members Debate Capacity Reduction: The Key is Precision." *China News*, https://www.chinanews.com.cn/m/gn/2016/03-02/7781298.shtml.

Lu, Zhiyao. 2016. "State of Play in the Chinese Steel Industry." https://www.piie.com/blogs/china-economic-watch/state-play-chinese-steel-industry.

MacKay, Alexander, and Nathan Miller. 2024. "Estimating models of supply and demand: Instruments and covariance restrictions." *American Economic Journal: Microeconomics*. Forthcoming.

Ma, Jinlong. 2011. "On-grid electricity tariffs in China: Development, reform and prospects." *Energy Policy*, 39(5): 2633–2645.

Matschke, Xenia, and Shane M. Sherlund. 2006. "Do Labor Issues Matter in the Determination of U.S. Trade Policy? An Empirical Reevaluation." *American Economic Review*, 96(1): 405–421.

McCalman, Phillip. 2004. "Protection for Sale and Trade Liberalization: An Empirical Investigation." *Review of International Economics*, 12(1): 81–94.

McGrath, Matt. 2019. "Climate Change: China Coal Surge Threatens Paris Targets." *BBC News* https://www.bbc.com/news/science-environment-50474824.

Ministry of Ecology and Environment. 2020. "Implementation Plan for the 2019–2020 National Carbon Emission Trading Quota Setting and Allocation (Power Generation Industry)." English translation by China Energy Portal `https://chinaenergyportal.org/en/implementation-plan-for-the-2019-2020-national-carbon-emission-trading-quota-setting-`

Miravete, Eugenio J, María J Moral, and Jeff Thurk. 2018. "Fuel taxation, emissions policy, and competitive advantage in the diffusion of European diesel automobiles." *RAND Journal of Economics*, 49(3): 504–540.

Mitra, Devashish, Dimitrios D. Thomakos, and Mehmet A. Ulubaşoğlu. 2002. ""Protection for Sale" in a Developing Country: Democracy vs. Dictatorship." *Review of Economics and Statistics*, 84(3): 497–508.

Montgomery, W David. 1972. "Markets in licenses and efficient pollution control programs." *Journal of Economic Theory*, 5(3): 395–418.

Muller, Nicholas Z, and Robert Mendelsohn. 2009. "Efficient pollution regulation: getting the prices right." *American Economic Review*, 99(5): 1714–1739.

National Energy Report. 2020. "What Makes Anthracite Different from Ordinary Coal?" `https://www.nengyuanjie.net/article/46326.html`.

NDRC. 2016*a*. "Measures for the Guaranteed Full Purchase of Renewable Electricity." *Notice of the National Development and Reform Commission on Issuing the Measures for the Guaranteed Full Purchase of Renewable Electricity* `https://www.ndrc.gov.cn/xxgk/zcfb/tz/201603/t20160328_963621.html`.

NDRC. 2016*b*. "NDRC Introduces Measures to Encourage Signing of Medium and Long-Term Contracts to Promote Sustainable Development in the Coal and Related Industries." `https://www.gov.cn/xinwen/2016-11/09/content_5130581.htm`.

NDRC. 2017. "Notice on Promoting the Signing and Fulfillment of 2018 Medium- and Long-Term Coal Contracts." National Development and Reform Commission `https://www.ndrc.gov.cn/xxgk/zcfb/tz/201711/t20171129_962607.html`.

NDRC. 2019. "Key Tasks for Resolving Overcapacity in the Coal Sector in 2019." National Development and Reform Commission of the People's Republic of China, Beijing `https://www.gov.cn/xinwen/2019-05/09/5390005/files/4fb50c72fd13411bb00e8693f948416f.pdf`.

NDRC. 2021*a*. "On Adjusting the Safety Stock Standards for Coal-Fired Power Plants in Peak Seasons." https://www.ndrc.gov.cn/fzggw/jgsj/yxj/sjdt/202109/t20210921_1297181.html.

NDRC. 2021*b*. "Shanxi: Resolutely Achieve the Task of Supplying 53 Million Tons of Coal for 14 Provinces and Cities in the Fourth Quarter." https://www.ndrc.gov.cn/wsdwhfz/202111/t20211105_1303073.html.

Newell, Richard G, and William A Pizer. 2003. "Regulating stock externalities under uncertainty." *Journal of Environmental Economics and Management*, 45(2): 416–432.

Newey, Whitney K, and James L Powell. 2003. "Instrumental variable estimation of nonparametric models." *Econometrica*, 71(5): 1565–1578.

Nie, Yu Marco. 2012. "Transaction costs and tradable mobility credits." *Transportation Research Part B: Methodological*, 46(1): 189–203.

People's Daily. 2016. "How to Ensure Capacity Cuts Are Real and Irreversible." https://www.nea.gov.cn/2016-10/26/c_135782310.htm.

Pigou, Arthur Cecil. 1920. *The Economics of Welfare.* London:Macmillan.

Pindyck, Robert S. 2013. "Climate change policy: what do the models tell us?" *Journal of Economic Literature*, 51(3): 860–872.

Pizer, William A. 2002. "Combining price and quantity controls to mitigate global climate change." *Journal of Public Economics*, 85(3): 409–434.

Qin, Ping, Xiao-Bing Zhang, and Lunyu Xie. 2020. "Coal taxation reform in China and its distributional effects on residential consumers." *Energy Policy*, 139: 111366.

Quirk, James P., and Rubin Saposnik. 1962. "Admissibility and Measurable Utility Functions." *Review of Economic Studies*, 29(2): 140–146.

Reuters. 2016. "China eyes $724 billion of transport investment over next three years." https://www.reuters.com/article/business/china-eyes-724-billion-of-transport-investment-over-next-three-years-idUSKCN0Y304Z/.

Roberts, Marc J., and Michael Spence. 1976. "Effluent Charges and Licenses under Uncertainty." *Journal of Public Economics*, 5(3–4): 193–208.

Rotemberg, Martin. 2019. "Equilibrium effects of firm subsidies." *American Economic Review*, 109(10): 3475–3513.

Rubens, Michael. 2022. "Oligopsony power and factor-biased technology adoption." National Bureau of Economic Research.

Ryan, Stephen P. 2012. "The costs of environmental regulation in a concentrated industry." *Econometrica*, 80(3): 1019–1061.

Shanghai Securities News. 2014. "China News: Shenhua Coal's Annual Pricing Strategy and Market Impact." https://www.chinanews.com.cn/ny/2014/12-30/6922742.shtml.

Shanghai Securities News. 2015. "Guizhou Financial News Reports: Insights into Shenhua Coal Pricing Adjustments." http://www.gzjrw.com.cn/Item/260032.aspx.

Shanxi. 2015. "Notice on Issuing the Several Measures of Shanxi Province to Reduce Enterprise Burdens and Promote Stable Industrial Operation." *Provincial government policy document*, https://www.shanxi.gov.cn/zfxxgk/zfxxgkzl/fdzdgknr/lzyj/szfwj/202205/t20220513_5976050.shtml.

Shanxi. 2016. "Report on the Execution of Shanxi Province's 2015 Budget and the Draft Budget for 2016." *Provincial budget report, hosted by the Ministry of Finance of the People's Republic of China*, https://www.mof.gov.cn/zhuantihuigu/2016hb/201602/t20160218_1760497.htm.

Shi, Xunpeng, Bertrand Rioux, and Philipp Galkin. 2018. "Unintended consequences of China's coal capacity cut policy." *Energy Policy*, 113: 478–486.

Sider, Hal. 1983. "Safety and productivity in underground coal mining." *Review of Economics and Statistics*, 225–233.

STA. 2014. "Notice on Implementing the Coal Resource Tax Reform." *Caishui [2014] No.˜72*, https://fgk.chinatax.gov.cn/zcfgk/c102416/c5204018/content.html.

State Council. 2013. "The Guidance on Deepening the Reform of the Coal Market." https://www.gov.cn/gongbao/content/2013/content_2307043.htm.

State Council. 2014. "Energy Development Strategy Action Plan (2014–2020)." *Notice of the General Office of the State Council on Issuing the Energy Development Strategy Action Plan (2014–2020)* https://www.gov.cn/zhengce/content/2014-11/19/content_9222.htm.

Stavins, Robert N, and Gernot Wagner. 2022. "Martin Weitzman: A Gift That Keeps on Giving." *Journal of the Association of Environmental and Resource Economists*, 9(5): 843–850.

Teng, Meixuan, Paul J Burke, and Hua Liao. 2019. "The demand for coal among China's rural households: Estimates of price and income elasticities." *Energy Economics*, 80: 928–936.

Wang, Lu. 2015. "Annual Coal-Power Negotiations Begin: Shenhua Proposes Long-Term Contract Price of RMB 400 per Ton." *Economic Information Daily*, https://energy.people.com.cn/n1/2015/1224/c71661-27969516.html.

Wang, Tongjun. 2021. "The intelligent Beijing–Zhangjiakou high-speed railway." *Engineering*, 7(12): 1665–1672.

Watson, Brett, Ian Lange, and Joshua Linn. 2023. "Coal demand, market forces, and US coal mine closures." *Economic Inquiry*, 61(1): 35–57.

Weitzman, Martin L. 1974. "Prices vs. Quantities." *Review of Economic Studies*, 41(4): 477–491.

Weitzman, Martin L. 2020. "Prices or quantities can dominate banking and borrowing." *Scandinavian Journal of Economics*, 122(2): 437–463.

Williams III, Roberton C. 2002. "Prices vs. Quantities vs. Tradable Quantities." National Bureau of Economic Research 9283.

Wright, Philip Green. 1928. *The tariff on animal and vegetable oils.* Macmillan.

Wright, Tim. 2012. *The political economy of the Chinese coal industry: black gold and blood-stained coal.* Routledge.

WSJ. 2022. "China's Coal Power Boom." *Wall Street Journal*, https://www.wsj.com/articles/chinas-coal-power-boom-beijing-xi-jinping-climate-energy-biden-administratio

Xinhua. 2014. "Coal tax reform to ease burden on producers, environment." *China Daily*, https://www.chinadaily.com.cn/business/2014-12/01/content_19002747.htm.

Xinhua. 2016. "Ten Coal Enterprises Sign Medium- and Long-Term Contracts." https://www.gov.cn/xinwen/2016-12/02/content_5141660.htm.

Xinhua. 2018. "China National Coal Group Signs Medium- and Long-Term Coal Supply Agreements with Six Major Power Central Enterprises." https://www.nea.gov.cn/2018-11/19/c_137617406.htm.

Xu, Shaofeng, and Wenying Chen. 2006. "The reform of electricity power sector in the PR of China." *Energy Policy*, 34(16): 2455–2465.

Yang, Yongqi, Ming Zeng, Song Xue, Jinyu Wang, and Yuanfei Li. 2018. "Unifying the "dual-track" pricing mechanism for coal in China: policy description, influences, and suggestions for government and generation enterprises." *Resources, Conservation and Recycling*, 129: 402–415.

Yi, Fujin, C-YC Lin Lawell, and Karen E Thome. 2019. "A Dynamic Model of Subsidies: Theory and Application to Ethanol Industry." Working paper, Cornell University.

Yuan, Jiahai, Xiaowen Yang, Ryna Yiyun Cui, Haonan Zhang, Fang Fang, Jian Zhang, and Yufei Ren. 2025. "The social impacts on China's vulnerable cities in the transition away from coal power." *iScience*, 28(11).

Yuan, Jingzhu. 2025. "Achievements of Coal Energy Development During the 14th Five-Year Plan Period and Prospects for the 15th Five-Year Plan Period." *Natural Resource Economics of China*, (4).

Zhang, Xiang, Lang Hong, and Ling Wu. 2020. "How to Improve the Long-Term Contract System to Stabilize the Coal Market?" *China Coal News*, `https://news.sina.com.cn/o/2020-12-10/doc-iiznctke5799736.shtml`.

Zhao, Chunyan. 2014. "Sales Decline for Four Consecutive Months: Shenhua Coal Prices Cut to an Eight-Year Low." *National Business Daily*, `https://www.nbd.com.cn/articles/2014-06-27/844707.html`.

Zhao, Chunyan. 2015. "Annual Coal-Power Negotiations Finalized: Power Companies Accept Shenhua's RMB 520 per Ton Long-Term Contract Price." *National Business Daily*, `https://www.nbd.com.cn/articles/2015-01-30/894551.html`.

Zhao, Xiaoli, and Chunbo Ma. 2013. "Deregulation, vertical unbundling and the performance of China's large coal-fired power plants." *Energy Economics*, 40: 474–483.

Zhao, Ziyuan. 2021. "How Can Coal Power Plants Efficiently Utilize Mixed Fuels?" *International Energy Network*, `https://m.in-en.com/article/html/energy-2310645.shtml`.

Zheng, Honghao. 2024. "Death by Market Power and the Production-Safety Tradeoff in the Coal Mining Industry." Working paper.

Zhou, Lin, Jianglong Li, Yangqing Dan, Chunping Xie, Houyin Long, and Hongxun Liu. 2019. "Entering and exiting: Productivity evolution of energy supply in China." *Sustainability*, 11(4): 983.

A Policy Documents

Figure 6 provides a snapshot of the original government document for the supply reduction policy. We provide an English translation of this document in this section.

In order to implement the *"Opinions of the State Council on Resolving Overcapacity in the Coal Industry and Achieving Sustainable Development"* (Guo Fa [2016] No. 7, hereinafter referred to as the "Opinions"), and to further regulate and improve the coal production and management order, effectively resolve overcapacity, and promote the sustainable development of coal enterprises, the following notice is issued regarding related issues:

1. **Fully understand the importance of standardizing and improving the order of coal production and operation**

   Currently, the coal industry is in a serious predicament. Especially since last year, some coal enterprises have resorted to measures such as overcapacity production, compensating for price through volume, and low-price dumping to maintain production operations in an effort to seize market share. These actions have not only severely disrupted the normal order of coal production and operation but have also exacerbated the imbalance between supply and demand in the market, intensifying the difficulties faced by the industry. If these issues are not promptly curbed, the national goals and tasks related to the development of the coal industry will be difficult to achieve. All regions and enterprises must adopt a strategic perspective that promotes supply-side structural reform and effectively addresses capacity reduction tasks. It is essential to enhance overall awareness and fully recognize the importance of standardizing and improving the order of coal production and operation for facilitating the recovery and development of the coal industry. This requires unifying thought and action with the spirit of the "Opinions", taking effective measures to ensure the implementation of regulations concerning the control of overcapacity production, reduction of production volume, suspension of production during holidays, and maintenance of fair competition.

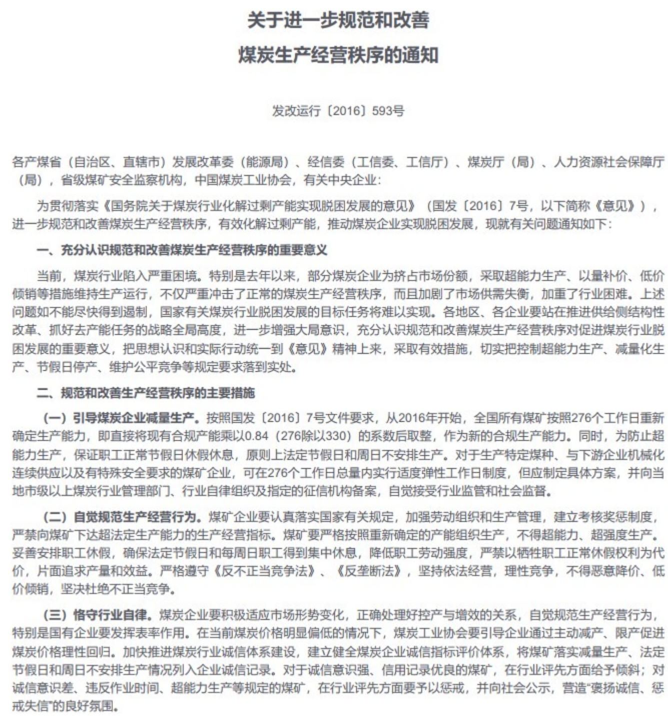
2. **Main Measures to Standardize and Improve Production and Operation Order**

   (a) **Guiding Coal Enterprises to Reduce Production Volume:** In accordance with the requirements of the State Council Document [2016] No. 7, starting from 2016, all coal mines nationwide are to re-determine their production capacity based on 276 working days. This means directly multiplying the existing compliant capacity by a coefficient of 0.84 (276 divided by 330) and rounding to the nearest whole number to establish the new compliant production capacity. Additionally, to prevent overcapacity production and ensure employees can take their normal holidays, no production should be scheduled on statutory holidays and Sundays. For coal mines producing specific coal types, providing mechanized continuous supply to downstream enterprises, or with special safety requirements, a moderately flexible working day system may be implemented within the total of 276 working days, provided that a specific plan is developed and filed with local coal industry management departments at or above the municipal level, industry self-regulatory organizations, and designated credit institutions, and that these enterprises consciously accept industry regulation and social supervision.

(b) **Consciously Standardizing Production and Operation Behavior:** Coal mines must earnestly implement national regulations, strengthen labor organization and production management, and establish a performance assessment and reward-punishment system. It is strictly prohibited to set production and operation targets that exceed the legal production capacity for coal mines. Production must be organized strictly according to the re-determined capacity, and overcapacity or excessively intense production is not allowed. Employee leave must be properly arranged to ensure that staff can rest during statutory holidays and every Sunday, thereby reducing labor intensity. It is forbidden to sacrifice employees' normal leave rights in order to pursue output and efficiency. Coal mines must strictly adhere to the "Anti-Unfair Competition Law" and the "Anti-Monopoly Law," operate in accordance with the law, engage in rational competition, refrain from malicious price cuts, and eliminate unfair competition.

(c) **Upholding Industry Self-Regulation:** Coal enterprises must actively adapt to changes in market conditions, correctly balance production control and efficiency enhancement, and consciously regulate their production and operation behavior, especially state-owned enterprises which should set a leading example. In the current context of notably low coal prices, the coal industry association should guide enterprises to promote rational price recovery through proactive production reduction and output limitations. The construction of an integrity system in the coal industry should be accelerated, establishing and improving an integrity index evaluation system for coal enterprises. Compliance with production reduction, as well as not scheduling production on statutory holidays and Sundays, should be included in the integrity records of enterprises. Coal mines with strong integrity awareness and good credit records should receive favorable treatment in industry evaluations, while those with poor integrity awareness, violations of operational hours, and overcapacity production should face penalties in industry evaluations and be publicly disclosed to create a positive atmosphere of "rewarding integrity and punishing dishonesty."

Figure 6: Policy Document: Supply Reduction Policy



关于进一步规范和改善
煤炭生产经营秩序的通知

发改运行〔2016〕593号

各产煤省（自治区、直辖市）发展改革委（能源局）、经信委（工信委、工信厅）、煤炭厅（局）、人力资源社会保障厅（局），省级煤矿安全监察机构，中国煤炭工业协会，有关中央企业：

为贯彻落实《国务院关于煤炭行业化解过剩产能实现脱困发展的意见》（国发〔2016〕7号，以下简称《意见》），进一步规范和改善煤炭生产经营秩序，有效化解过剩产能，推动煤炭企业实现脱困发展，现就有关问题通知如下：

一、充分认识规范和改善煤炭生产经营秩序的重要意义

当前，煤炭行业陷入严重困境。特别是去年以来，部分煤炭企业为挤占市场份额，采取超能力生产、以量补价、低价倾销等措施维持生产运行，不仅严重冲击了正常的煤炭生产经营秩序，而且加剧了市场供需失衡，加重了行业困难。上述问题如不能尽快得到遏制，国家有关煤炭行业脱困发展的目标任务将难以实现。各地区、各企业要站在推进供给侧结构性改革、抓好去产能任务的战略全局高度，进一步增强大局意识，充分认识规范和改善煤炭生产经营秩序对促进煤炭行业脱困发展的重要意义，把思想认识和实际行动统一到《意见》精神上来，采取有效措施，切实把控制超能力生产、减量化生产、节假日停产、维护公平竞争等规定要求落到实处。

二、规范和改善生产经营秩序的主要措施

（一）引导煤炭企业减量生产。按照国发〔2016〕7号文件要求，从2016年开始，全国所有煤矿按照276个工作日重新确定生产能力，即直接将现有合规产能乘以0.84（276除以330）的系数后取整，作为新的合规生产能力。同时，为防止超能力生产，保证职工正常节假日休假休息，原则上法定节假日和周日不安排生产。对于生产特定煤种、与下游企业机械化连续供应以及有特殊安全要求的煤矿企业，可在276个工作日总量内实行适度弹性工作日制度，但应制定具体方案，并向当地市级以上煤炭行业管理部门、行业自律组织及指定的征信机构备案，自觉接受行业监管和社会监督。

（二）自觉规范生产经营行为。煤矿企业要认真落实国家有关规定，加强劳动组织和生产管理，建立考核奖惩制度，严禁向煤矿下达超法定生产能力的生产经营指标。煤矿要严格按照重新确定的产能组织生产，不得超能力、超强度生产。妥善安排职工休假，确保法定节假日和每周日职工得到集中休息，降低职工劳动强度，严禁以牺牲职工正常休假权利为代价，片面追求产量和效益。严格遵守《反不正当竞争法》、《反垄断法》，坚持依法经营，理性竞争，不得恶意降价、低价倾销，坚决杜绝不正当竞争。

（三）恪守行业自律。煤炭企业要积极适应市场形势变化，正确处理好控产与增效的关系，自觉规范生产经营行为，特别是国有企业要发挥表率作用。在当前煤炭价格明显偏低的情况下，煤炭工业协会要引导企业通过主动减产、限产促进煤炭价格理性回归。加快推进煤炭行业诚信体系建设，建立健全煤炭企业诚信指标评价体系，将煤矿落实减量生产、法定节假日和周日不安排生产情况列入企业诚信记录。对于诚信意识强、信用记录优良的煤矿，在行业评先方面给予倾斜；对诚信意识差、违反作业时间、超能力生产等规定的煤矿，在行业评先方面要予以惩戒，并向社会公示，营造"褒扬诚信、惩戒失信"的良好氛围。

B Data Comparison

To validate the provincial and monthly production and consumption data we compiled from various sources, we compare the statistics based on our data and those from the National Bureau of Statistics of China in Table B.1. The first column displays the national values aggregated from our data based on coal transportation, the second column is derived from the NBS Coal Balance Table for all provinces, and the third column represents the national statistics in the NBS National Coal Balance Table.

Overall, the national statistics derived from our transportation data are reasonably close to those in NBS. For instance, both datasets show declining output and consumption between 2014 and 2016, and a decrease in imports in 2015 followed by a rebound in 2016.

We also validate our data at the province-year level. Figure B.1 compares the shipment out of and into a province in each year with the annual coal production and consumption data by province in the provincial coal balance tables from NBS. Overall, they align closely, with the imputed consumption data being slightly lower than those reported by the NBS. This difference is expected because NBS data explicitly state that they do not remove certain

Table B.1: Data Comparison of Coal Output and Consumption (million tons)

	2014			2015			2016		
	Main	NBS (Prov)	NBS (Nat)	Main	NBS (Prov)	NBS (Nat)	Main	NBS (Prov)	NBS (Nat)
Output	3703	3876	3874	3457	3747	3747	3135	3411	3411
Consumption	3860	4317	4136	3662	4255	3998	3574	4249	3888
Thermal power	1974	1823	1895	1872	1792	1796	1908	1818	1827
Heating supply	181	306	224	204	323	241	222	345	266
Coking Coal	563	667	639	530	625	609	536	626	606

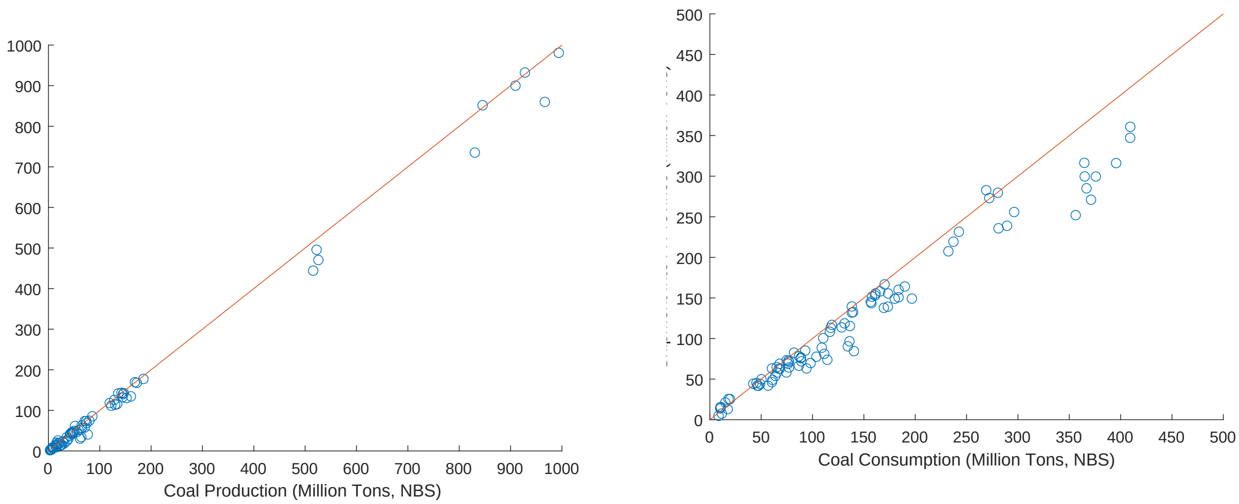
Notes: The table compares the transportation data used in the empirical analysis (under main columns) with the annual numbers reported by the National Bureau of Statistics of China. The column of NBS (Prov) contains data collected from province coal balances tables and aggregated to national numbers, and the column NBS (Nat) uses statistics reported in the national coal balance tables. The sum of coal consumption across provinces exceeds the national total because of the potential double counting according to the official Q&A from China NBS (See `https://www.stats.gov.cn/hd/cjwtjd/202302/t20230207_1902276.html`). Under the main column, the output is calculated as total shipment, and the monthly consumption is constructed in Appendix C. We then aggregate province-month production and consumption to the national level.

instances of double-counting at the province level.[40]

Figure B.1: Data Comparison of Shipment, Coal Output and Consumption at Province Levels (Million Tons)

(A) Shipment Out of a Province vs Production

(B) Shipment into a Province vs Consumption



The figure compares province-year coal production and consumption in the data used in our empirical analysis with those reported by province coal balance tables. One dot in the figures represents one province-year pair. The y-axis is calculated by aggregating monthly total shipment out of and into a province within a year.

C Coal Consumption

We collect monthly coal consumption data from 2014 to 2016 across seven sectors: electricity, coking, construction, heating supply, steel-making, chemistry, and others. Table C.1 displays annual consumption in these sectors. Electricity is the largest coal consumer, accounting for over 50% of total consumption. Other major users include coking and cement plants, each accounting for 15%. The consumption breakdown is stable during our sample period. The "others" sector, which mainly includes residential demand and small-scale industrial processes, accounted for 7% in 2014 and declined to 3% in 2016. We plot the national time series of the downstream output in Figure C.1.

[40]For example, coal transported to province A for washing and burned at power plants in B are both counted as consumption in provinces A and B. Our shipment data only count the final consumption in B.

Table C.1: Consumption Shares by Sectors

Year	2014	2015	2016
Electricity	0.51	0.51	0.53
Coking	0.15	0.14	0.15
Construction	0.15	0.15	0.14
Heating Supply	0.05	0.06	0.06
Steel-making (steam coal only)	0.04	0.04	0.04
Chemistry	0.04	0.04	0.04
Others	0.07	0.06	0.03

Note: The table shows annual coal consumption shares by sector. We collected monthly coal consumption by sector from China Coal Resource, and then aggregated them annually to calculate those shares.

To allocate coal consumption by sector across provinces, we further collect the monthly output of primary products in each sector by province from the NBS and convert it into coal consumption. For the electricity sector, we use thermal electricity output as the main product. Although thermal electricity generation includes both coal and natural gas, natural gas contributes to only about 4.5% of total thermal generation, making its impact negligible.[41] We use cement as the representative product for the construction sector, as it accounts for 83.7% of coal consumption there. In the chemistry sector, fertilizers serve as the primary product.

Provincial coal balance tables provide annual coal consumption for thermal electricity generation and coking. We aggregate thermal and coke output to the province-year level and calculate coal consumption per 1 MWh of electricity and per ton of coke for each province and year. We assume that these ratios remain constant across months within a province and year, converting thermal electricity generation and coke into coal consumption for the electricity and coking sectors. For cement, steel, and fertilizers, we aggregate their output by month across provinces and calculate the coal-to-output ratio. Assuming this ratio is constant across provinces within a month, we compute coal consumption in the construction, steel, and chemistry sectors based on their province-month output. Table C.2 shows that producing 1 MWh of electricity uses less coal over time, reflecting the gradual improvement in power plant energy efficiency. The conversion ratios are also consistent with engineering estimates by industry experts.[42]

[41] https://chinaenergyportal.org/en/2016-detailed-electricity-statistics-updated/

[42] For example, producing 1 ton of cement requires 0.11 ton of standard coal, equivalent to a conversion ratio of 0.15 between coal and cement, given that 1 ton of raw coal equals 0.7143 ton of standard coal. See https://www.chinanews.com.cn/cj/2011/05-03/3011403.shtml.

Table C.2: Coal-to-End Product Output Ratio by Sectors

Year	2014	2015	2016
Electricity (Ton of coal/MWh)	0.47	0.44	0.43
Cement (Ton of coal/Ton)	0.23	0.24	0.21
Steel (Ton of coal/Ton)	0.14	0.12	0.13
Coke (Ton of coal/Ton)	1.18	1.19	1.20
Fertilizer (Ton of coal/Ton)	2.01	1.85	2.00

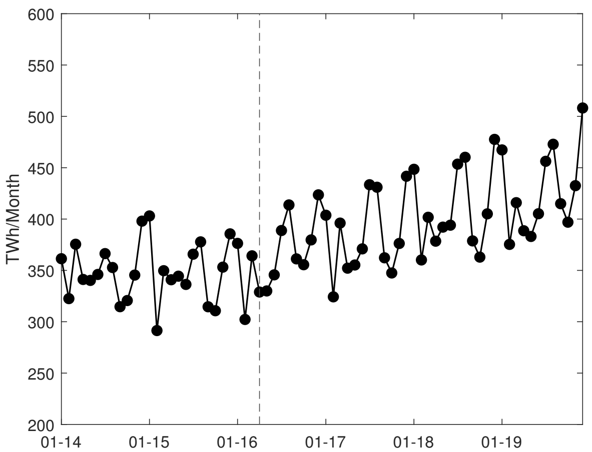
Note: The table reports the conversion ratios used to estimate coal consumption for the production of key downstream products in each sector. Monthly data on the output of thermal electricity, cement, steel, coke, and fertilizers by province is collected from the NBS. The provincial coal balance tables provide annual coal consumption for thermal electricity generation and coking. We aggregate thermal and coke output to the province-year level to calculate the coal consumption per 1 MWh of electricity and per ton of coke. The table shows the average across provinces for each year. For cement, steel, and fertilizers, we calculate national output annually and match it with coal consumption data for the construction, steel (steam coal only), and chemical sectors from China Coal Resource, resulting in an annual conversion rate applied uniformly across all provinces.

For the heating sector and the "other" sector, which lack representative products, we collect annual coal consumption for heating supply from provincial coal balance tables. We then compute annual consumption in the "other" sector as the residual of total coal consumption after accounting for electricity generation, heating supply, industry, and construction. Using these data, we first calculate province-specific annual coal consumption shares by sector and then allocate monthly consumption across sectors according to these shares.

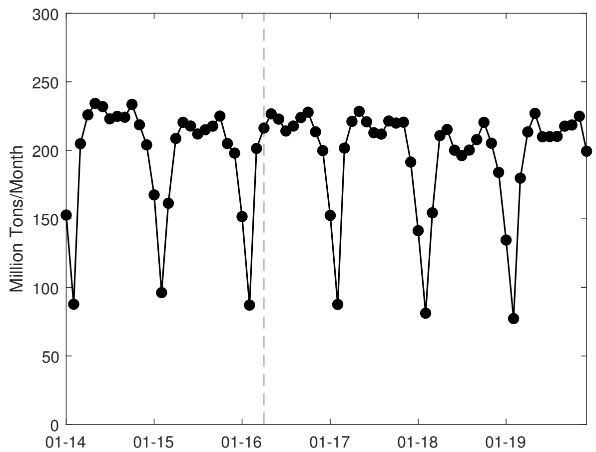
We extrapolate these data to 2017. Our monthly, sector-level consumption data and cross-province shipment data end in 2016. For 2017, we impute the consumption data using the province-level monthly production data from NBS. Specifically, we allocate each province's production to destination provinces using shipment shares observed in 2016. We note that the shipment quantity is slightly lower than NBS's production data in prior years, likely due to a double-counting issue with the NBS data as discussed in Appendix B. Therefore, the allocated production is scaled down by 0.936, the average production-to-shipment ratio observed in 2014-15. We assume that the total production allocated to a province is its consumption, which is a reasonable assumption given that coal inventory has stabilized in 2017 (Figure 2).

Figure C.1: Output in Major Downstream Sectors

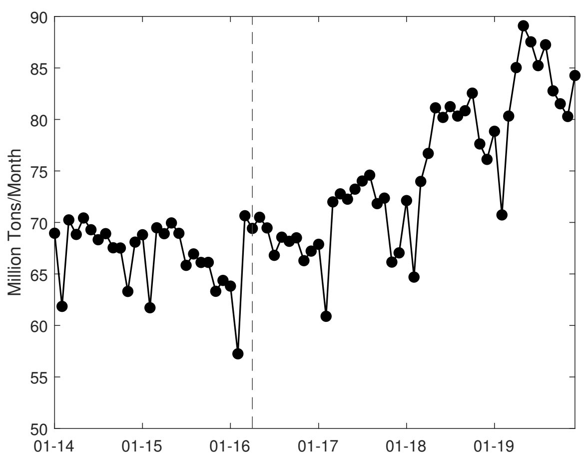
(a): Thermal Electricity



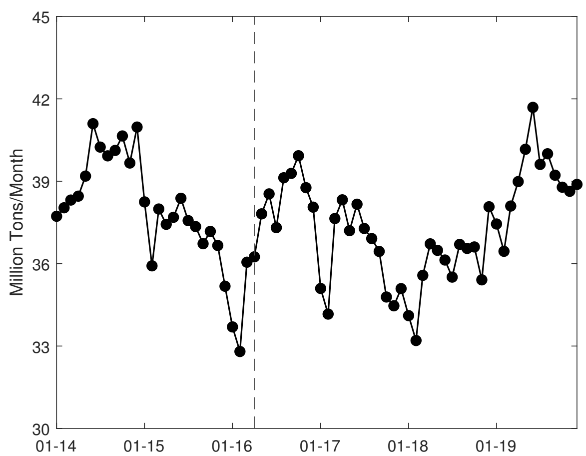
(b): Cement



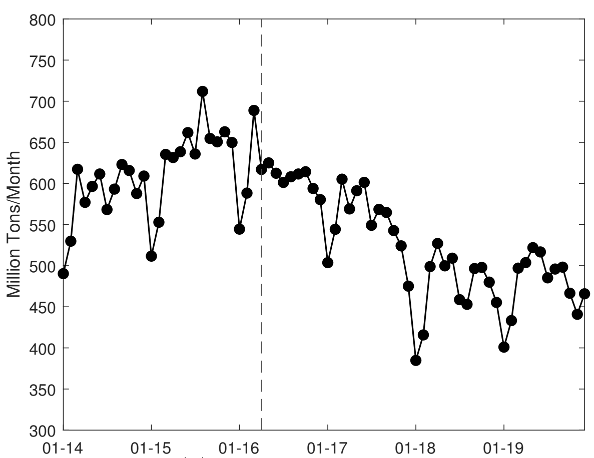
(c): Steel



(d): Coke



(e): Fertilizer



Notes: Panel (a) plots the national thermal electricity generation, (b) plots the national cement production, (c) plots the national steel production, (d) plots the national coke production, (e) plots the national fertilizer production.

D China's Major Infrastructure Development in April and May of 2016

The list below represents our best-effort search for major active infrastructure projects (with a cost over RMB 50 billion) in April and May of 2016, based on NDRC announcements, Xinhua News, China Daily and other news sources. We focus on projects that are extremely material-intensive, thus requiring significant use of coal and coke for concrete and steel. We exclude the Guangzhou–Shenzhen–Hong Kong Express Rail Link (Hong Kong section), which meets the RMB 50 billion cost threshold, but the scale (26km of railway) was far smaller than the other projects.

New Projects

1. Yuxi–Mohan Railway (China-Laos Railway) in Yunnan (southwest of China). Full-scale construction began on April 19, 2016, following earlier preparatory works that started in September 2015. The project's total cost is about RMB 51.6 billion.[43]

2. Beijing–Zhangjiakou High-Speed Railway (HSR) in Beijing and Hebei (central China). The construction started on April 29th, 2016 (Wang, 2021), with a total investment of RMB 58.4 billion.[44]

3. Tianfu International Airport in Chengdu (southwest of China). The construction started in May 2016, with a total investment of RMB 72 billion.[45]

Ongoing Projects

1. Haoji Railway between Shanxi and Inner Mongolia (northwest-central China). The construction started in 2015 and was completed in 2019, with a cost more than RMB 193 billion.[46]

2. Daxing International Airport in Beijing and Hebei (central China). The construction started in 2014 and was completed in 2019, with a cost of RMB 80 billion.[47]

[43] https://global.chinadaily.com.cn/a/202007/03/WS5efe84baa310834817256ef1.html

[44] https://www.reuters.com/article/sports/china-approves-9-2-billion-winter-olympics-high-speed-rail-line-local-government-idUSKCN0SS04F/

[45] https://aviationweek.com/air-transport/chinas-new-chengdu-airport-facilitates-massive-air-travel-growth

[46] https://global.chinadaily.com.cn/a/201910/05/WS5d97d87fa310cf3e3556ecde.html

[47] https://www.chinadaily.com.cn/business/2014-12/26/content_19177195.htm

3. Beijing–Shenyang HSR in Beijing, Hebei and Liaoning (central-north China). The construction began in 2014 and was completed in 2021, with a cost of RMB 123.5 billion.[48]

4. Hong Kong–Zhuhai–Macau Bridge in Guangdong, Hong Kong and Macau (south China). The construction began in 2009 and was completed in 2018, with a cost of RMB 127 billion.[49]

5. Wudongde Hydropower Station in Sichuan and Yunnan (southwest of China). The construction started in 2015 and was completed in 2021, with a cost of RMB 120 billion.[50]

6. Lianghekou Hydropower Station in Sichuan (southwest of China). The construction started in 2014 and was completed in 2022, with a cost of RMB 66.5 billion.[51]

E Coal Transportation

An integrated railroad and coastal marine shipping system is the primary mode of coal transportation in China, accounting for 65.2% of total coal transport in 2017.[52] The transportation costs typically include the direct expenses associated with freight charges, which are influenced by the distance to be traveled, the mode of transportation (such as rail, truck, or barge), and the specific logistics involved. While shipping fares for waterway transport are influenced by market forces, railroad transportation fares are typically regulated.[53] Additionally, loading and unloading fees, terminal charges, and any necessary handling costs contribute to the total transportation expenditure. Other factors, such as fuel prices, maintenance of transportation vehicles, and labor costs, also play a significant role in shaping these costs.

Due to the limited variation in transportation shares, directly estimating transportation costs through shipping choices via a behavioral model is challenging. Instead, we manually collected estimated transportation costs between major coal producers and consumers from an authoritative industry report (Cao and Feng, 2017) published by China Bond Rating Co.,

[48]https://usa.chinadaily.com.cn/china/2013-12/30/content_17205381.htm

[49]https://web.archive.org/web/20190327195326/http://www.xinhuanet.com/english/2018-10/23/c_137553194.htm

[50]https://www.reuters.com/business/energy/chinas-giant-wudongde-hydro-project-put-into-full-operation-2021-06-16/

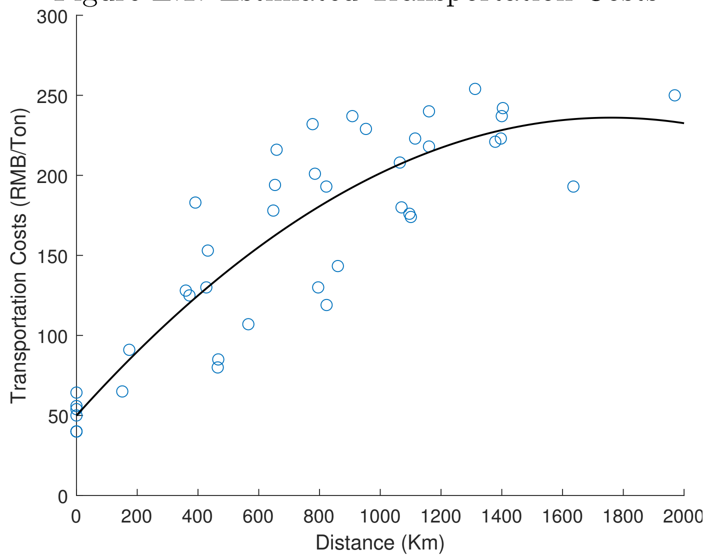
[51]https://www.chinadailyhk.com/hk/article/264174

[52]See http://pdf.dfcfw.com/pdf/H3_AP201811151246522585_1.PDF.

[53]For more details about the composition of regulated fares for transporting coal by railroad, please refer to http://www.gov.cn/xinwen/2017-12/26/5250421/files/5a71188c4fd34d1d9f05ef5e16690d2e.pdf.

Ltd., a leading agency that investigates and collects industry information to assess default risks of bonds across various sectors, including coal. Then we fit a quadratic relationship between the transportation costs and the distance between any two provinces, imposing the restriction that transportation costs increase with distance up to the maximum distance between any two provinces in China. Figure E.1 shows the transportation costs of province pairs estimated by the industry report, along with the estimated relationship between inter-provincial distance and transportation costs. The intercept represents the costs of transporting coal within a province, estimated to be RMB 60.9/ton. The estimates also indicate slight distance discounts.

Figure E.1: Estimated Transportation Costs



Notes: The figure shows transportation costs (RMB/ton) between major coal-producing and coal-consuming provinces, plotted against the distance between their capital cities (km), based on an industry report (Cao and Feng, 2017) published by China Bond Rating Co., Ltd. We then approximate the relationship between transportation costs and distance using a quadratic function (black curve).

F Simulation of Demand and Supply Shocks

We estimate the following AR(1) processes for demand shocks ε_{jt} and supply shocks ξ_{it}:

$$\varepsilon_{jt} = \delta_0^D + \delta_1^D \varepsilon_{jt-1} + \omega_{jt}^D$$

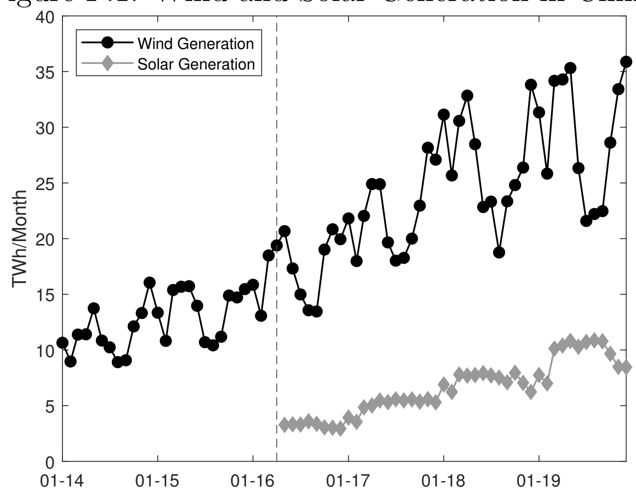
$$\xi_{it} = \delta_0^S + \delta_1^S \xi_{it-1} + \omega_{it}^S,$$

Table F.1 reports the estimated AR(1) coefficients, showing that demand shocks exhibit greater persistence than cost shocks.

	Demand	Domestic Supply
δ_1	0.805	0.494
	(0.025)	(0.066)
N	1020	850

Table F.1: Estimated AR(1) Process for Demand and Cost Shocks

Figure F.1: Wind and Solar Generation in China



Note: We plot the total monthly wind and solar generation in China. The solar generation data are available after April 2016, and their levels are considerably lower than the wind's.

To forecast future demand, we additionally model the evolution of wind generation and cooling degree days (CDD). Motivated by the growth pattern of the total wind generation at the national (Figure F.1) and province level, we model the generation as a province-specific linear time trend with province-month-of-the-year fixed effects and shocks:

$$X_{jt}^{\text{Wind}} = \delta_{jm(t)}^{\text{Wind}} + \delta_{j1}^{\text{Wind}} t + \omega_{jt}^{\text{Wind}},$$

which is estimated separately for each province. We assume that $\omega_{jt}^{D}, \omega_{it}^{S}, \omega_{jt}^{\text{Wind}}$ are mutually independent in the forward simulation. We re-sample historical CDDs by year, under the assumption that cross-year climate variation over the sample period is stable.

G Long-Term Contracts and Demand Elasticities

An important assumption in Section 5 is that the monthly coal price in a province is determined in the market equilibrium. In this section, we explore the robustness to this assumption by examining the sensitivity of demand elasticities when a share of demand is locked in by the contracts.

Specifically, we assume that long-term contracts, often signed at the beginning of a year, commit buyers and sellers to trading a fixed quantity at a fixed price for the coming year. This is the original intent of the long-term contracts, although, as Section 2 shows, few buyers and sellers were compliant in 2014 and 2016. To see how the potential contractual lock-in affects estimated demand elasticities, we consider the following robustness analyses: we assume that a share of $x_{jn}\%$ of observed quantity demanded in year n and province j, $Q^{\star}_{jn}$, is the contracted amount. This amount is purchased in equal amount over the 12 months. Therefore, the coal demand in the spot market in Equation (10) becomes

$$\ln(q_{jt}\left(p_{jt}\right) - \frac{Q^{\star}_{jn}}{12} x\%) = \alpha^{D} \ln p_{jt} + \beta_X X_{jt} + FE_{j}^{\text{demand}} + FE_{m(t)}^{\text{demand}} + \varepsilon_{jt} \tag{G.1}$$

For simplicity, we assume that the share is uniform across provinces $x_{jn} \equiv x_n$ and differs across years. In Table G.1, we report the estimates of the spot market elasticity α^D and the quantity-weighted average adjusted elasticity based on the total quantity for two sequences of x_n. In panel A, we assume that the fixed quantity share is 7% for 2014-2016, and is 50% for 2017. These estimates are based on the discussion in Section 2.[54] In panel B, we exclude 2017, the year most affected by the contracted coal and re-estimate. We find that the implied elasticities in both panels are smaller, but still statistically not distinguishable from the baseline estimates in Section 5.

[54]In reality, the locked-in amounts likely differ across months in a year, especially in 2014-2016, but it remains of interest whether our demand functional form is robust to the lock-in effects from such contracts.

Table G.1: Locked-Contract Demand: Quantity-Weighted Average Elasticities

Panel A: 2014–2017	Timing	SOE	Small-Mine Capacity	Timing+SOE
α^D	-0.747	-0.673	-1.353	-0.689
	(0.215)	(0.203)	(0.509)	(0.204)
Adj. avg. elasticity	-0.613	-0.552	-1.110	-0.565
	(0.176)	(0.166)	(0.418)	(0.167)
Panel B: 2014-2016	Timing	SOE	Small-Mine Capacity	Timing+SOE
α^D	-0.427	-0.391	-0.981	-0.404
	(0.192)	(0.195)	(0.402)	(0.192)
Adj. avg. elasticity	-0.398	-0.365	-0.914	-0.377
	(0.180)	(0.183)	(0.376)	(0.180)

Notes: The table reports locked-contract demand estimates for two samples. Panel A uses annual contract shares [0.074, 0.074, 0.074, 0.573] for 2014–2017. Panel B uses [0.074, 0.074, 0.074] for 2014–2016. Each estimate is from the log spot-demand regression using the indicated IV set. The adjusted average elasticity accounts for fixed quantities by scaling the spot-demand price coefficient by the spot-demand share in total demand for each province-month and then taking the quantity-weighted average across province-month observations. Standard errors for the adjusted average elasticity treat the fixed-quantity shares as fixed and apply the same quantity-weighting transformation to the coefficient standard error. All specifications exclude Beijing and include province and month fixed effects plus wind generation and cooling degree days.

H Coal Consumption and Pollution Damages

To quantify the environmental impact of reducing coal production, we additionally model air pollution as a function of coal consumption and health benefits from changes in pollution levels. We also include the damage from coal production based on water depletion.

H.1 Air Pollution

We specify a mapping from coal consumption to monthly PM2.5 and PM10 concentrations within a province,[55] which are the primary predictors of health costs in the literature we use later. We estimate the following model for the concentration of pollutant $\ell \in \{\text{PM2.5, PM10}\}$ in province j and month t (where the coefficients are ℓ-specific):

$$\ln \ell_{jt} = \psi_C \ln \tilde{C}_{jt} + \psi_X X_{jt} + FE_j^{\text{pollution}} + FE_{m(t)}^{\text{pollution}} + \omega_{jt}. \tag{H.1}$$

[55] We use the average of all monitoring stations' concentration readings in a province.

In the above, to account for spillover effects (Fu, Viard and Zhang, 2022), we define the variable $\tilde{C}_{jt}$ as the total coal consumption in province j and all of its neighboring provinces in period t.[56] The variable X_{jt} includes meteorological conditions, including temperature, dew point temperature, sea level pressure, wind speed, sky condition total coverage code, precipitation duration and depth. Additionally, we control for fixed effects at the province and month-of-the-year levels.

Coal consumption and the unobservable ω_{jt} may be correlated. Specifically, the coal consumption could be correlated with other economic activities that also contribute to pollution, but also correlated with heightened monitoring and regulation that reduces pollution. We therefore use the same IVs as the demand estimation for coal consumption.[57]

Table H.1 indicates that a 1% increase in coal consumption in a province and its neighbors raises local PM2.5 concentration by about 1.4%. The effect is similar when we use PM10 concentration as the outcome variable.[58]

Table H.1: The Effects of Coal Consumption on Air Pollutants

	(1) ln(PM2.5) OLS	(2) ln(PM10) OLS	(3) ln(PM2.5) IV	(4) ln(PM10) IV
$\ln \tilde{C}_{jt}$	0.416***	0.386***	1.438***	1.284***
	(0.115)	(0.096)	(0.332)	(0.314)
Province FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
Meteorological Controls	YES	YES	YES	YES
Observations	1,440	1,440	1,440	1,440
Adjusted R^2	0.654	0.605	0.554	0.490
Kleibergen-Paap F Statistic			35.13	35.13

Notes: The table reports estimates of the effects of coal consumption on PM2.5 and PM10 concentrations by OLS and IV. All estimates control for meteorological conditions, including air temperature, dew point, sea level pressure, wind speed, sky condition coverage, and liquid precipitation duration and depth. The instruments used are the same as those in estimating demand. Standard errors are clustered at the province level.

[56] We do not further normalize the variables by the province's size because both the pollution measure and the consumption are logged, and we control for province fixed effects.

[57] We construct instruments for $\tilde{C}$ by summing the instruments across neighboring provinces.

[58] Ito and Zhang (2020) estimate the elasticity of PM10 concentration with respect to coal usage to be 0.5 using a regression discontinuity design exploiting the Huai-River heating policy. This is similar to the OLS estimates from our province-month panel data, but smaller than the IV estimates.

H.2 Monetize Pollution Damages

We monetize morbidity and mortality costs associated with changes in ambient pollutant concentrations following Barwick et al. (2021).[59] Specifically, they estimate that a 10 $\mu g/m^3$ reduction in daily PM2.5 generates \$9.2 billion in annual morbidity benefits. They also quantify mortality costs using a back-of-the-envelope calculation based on the mortality estimates in Ebenstein et al. (2017) (with data from 2004 to 2012). This calculation implies mortality costs of approximately \$13.4 billion per year associated with a 10 $\mu g/m^3$ increase in PM10. We convert all estimates to a per-person per-month basis and express them in 2016 RMB.[60] In the simulation, we first predict the percentage changes of concentrations of both pollutants from coal consumption changes based on the marginal effects of coal consumption estimated using IV, and then translate the level changes to the total morbidity and mortality health costs per capita. The total changes are calculated by weighting the changes by the population for each province and then summing across provinces.

We account for the environmental costs of water depletion at RMB 86.6 per ton of coal production (Gu and Li, 2017).

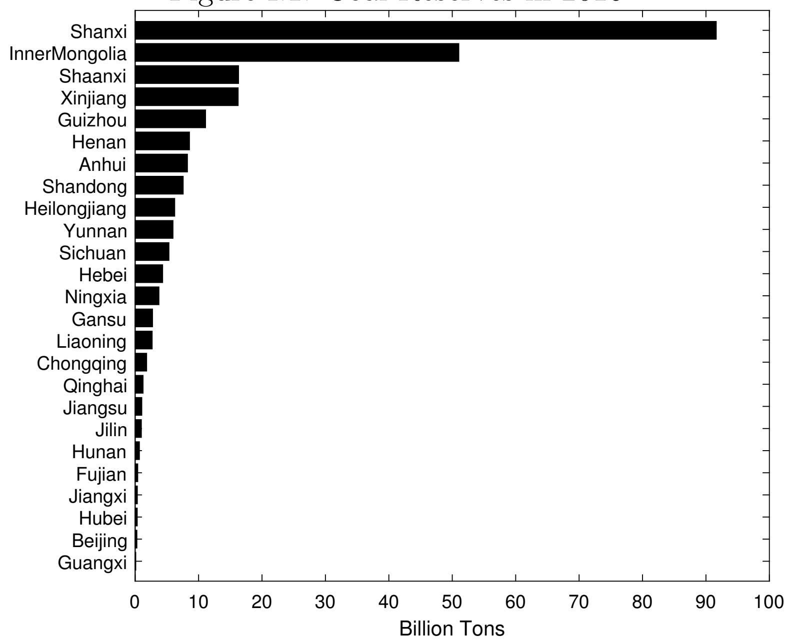
[59] An alternative is based on the approach in Ito and Zhang (2020) that estimate a household's willingness-to-pay of 1.34 dollars per year for 1 $\mu g/m^3$ of decrease in PM10 based on data from 2006 to 2014. They also derive an income-dependent willingness to pay based on their estimates of a random-coefficient logit model for air purifiers. This measure leads to a slightly lower monetary value for reducing air pollution.

[60] The health cost estimates reported in Barwick et al. (2021) are expressed in 2015 U.S. dollars. We convert these estimates to RMB using an exchange rate of RMB 6.5 per USD. To obtain per-capita values and express them in real 2016 terms, we use a national population of 1.375 billion and consumer price indices (CPI) of 615.2 in 2015 and 637.5 in 2016. Finally, we divide the resulting annual per-capita values by 12 to obtain monthly measures.

I Additional Figures and Tables

I.1 Distribution of Coal Reserves

Figure I.1: Coal Reserves in 2016

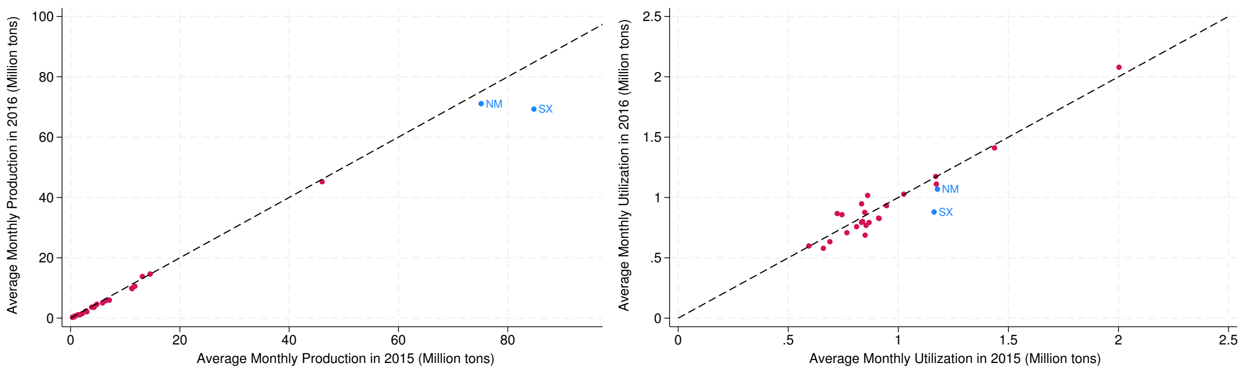


Notes: Reserves are coal resources that have been surveyed and are feasible for excavation based on data from Qianzhan Industrial Research Institute.

I.2 Policy Effects on Each Province and SOEs

Figure I.2: Shanxi and Inner Mongolia Reduce Production More in 2016

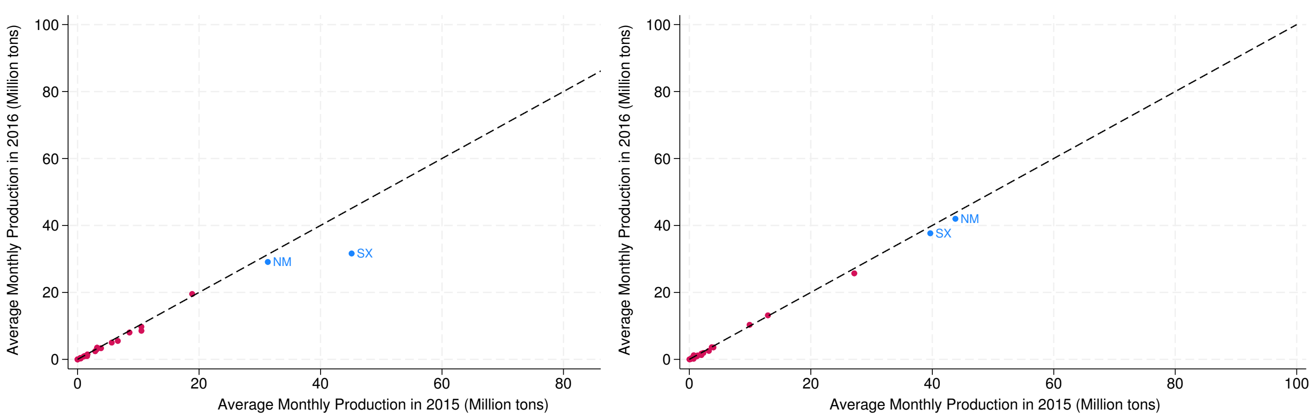
(a): Year-to-Year Production (b): Year-to-Year Utilization Factor



Notes: In panel (a), each dot represents a combination of 2015 (horizontal) and 2016 (vertical) average monthly production of a province. A location below the 45 degree line indicates that the 2016 production is lower. In panel (b), each dot represents a combination of average monthly utilization factors, defined as production over capacity (both normalized to the monthly level), of a province. A location below the 45 degree line indicates that the 2016 utilization factor is lower.

Figure I.3: SOE Reduces Production More in 2016

(a): Year-to-Year SOE Production (b): Year-to-Year Non-SOE Production



Notes: In panel (a), each dot represents a combination of 2015 (horizontal) and 2016 (vertical) average monthly production of a province's state-owned enterprises (SOE). A location below the 45 degree line indicates that the 2016 production is lower. In panel (b), each dot represents a combination of average monthly production by a province's non-SOE.

I.3 Concentration of Top State-owned Coal Companies

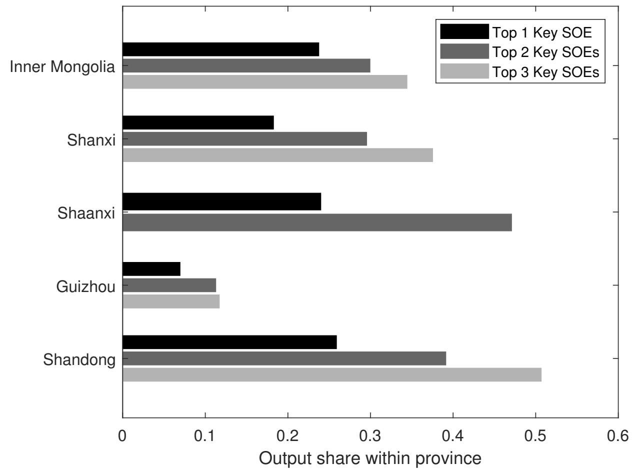
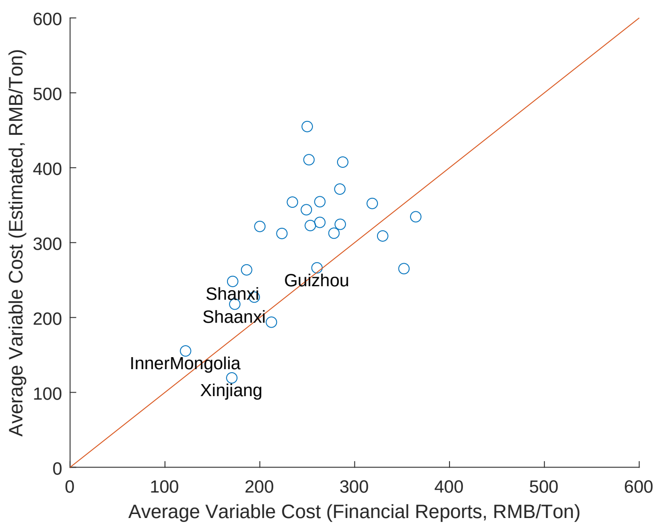


Figure I.4: Market Shares of Major SOE Coal Producers in Key Provinces

I.4 Comparison between Estimated Average Costs and Unit Costs from Financial Reports

Figure I.5: Compare Estimated Average Costs with Unit Costs from Financial Reports

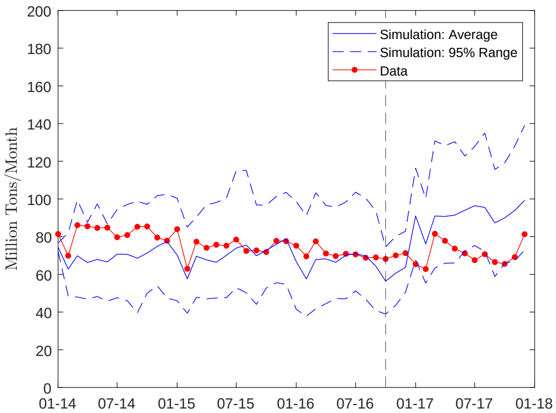


The figure compares average costs by province from the estimated model with unit costs from financial reports. The model's estimated average costs are calculated at the observed quantity and averaged between 2014 and 2016. The unit costs are collected from financial reports and bond prospectuses of major coal companies in each province, then averaged across all companies over three years in the sample.

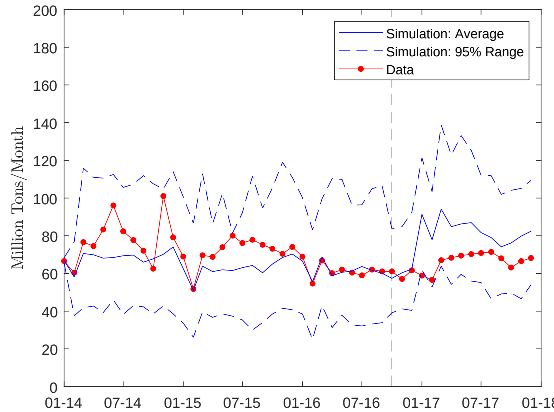
I.5 Model Fit by Provinces

Figure I.6: Simulations: Output by Provinces

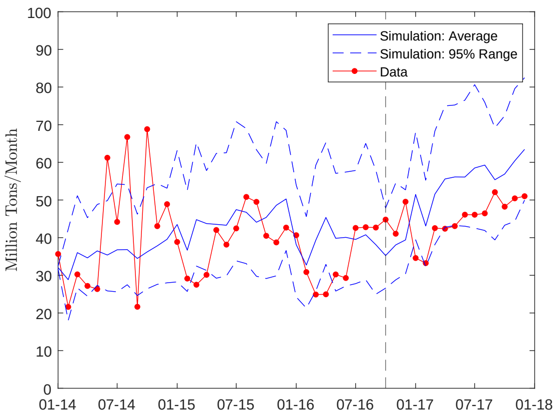
A: Inner Mongolia



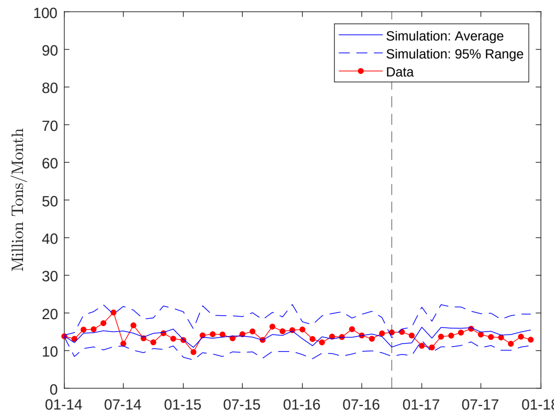
B: Shanxi



C: Shaanxi



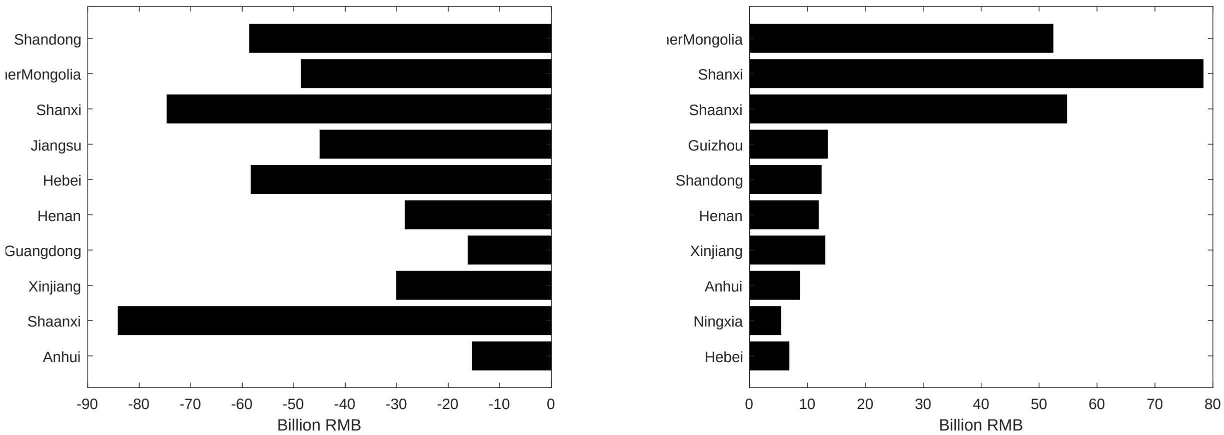
D: Guizhou



Figures plot the mean and 2.5%–97.5% range across all simulations for output in the four largest coal-producing provinces: Inner Mongolia, Shanxi, Shaanxi and Guizhou.

Figure I.7: Policy Effects on Economic Surpluses in Major Provinces

(a) Consumer Surplus (b) Producer Surplus



The figures illustrate changes in consumer surplus and producer surplus in 2017 relative to a no-policy counterfactual. Changes in consumer surplus are ordered by provinces' average pre-policy consumption, while changes in producer surplus are ordered by provinces' average pre-policy production.

I.6 Distribution of Economic Surplus by Top 10 Provinces

J Threshold Robustness

Table J.1: No-policy Effects: Threshold Robustness

	Lower Threshold	Higher Threshold
Δ Delivery price (RMB/ton)	-137	-158
	[-148, -126]	[-177, -139]
Δ Domestic coal production (million tons/year)	626	753
	[569, 697]	[628, 906]
Δ Coal consumption (million tons/year)	582	701
	[525, 652]	[581, 845]
Δ Welfare (RMB bn.)	-9	-101
	[-33, 15]	[-129, -74]

Table J.2: Policy Inefficiency Relative to Efficient Quotas: Threshold Robustness

	Lower Threshold		Higher Threshold	
	Within-Province	Total	Within-Province	Total
Δ Producer Surplus (RMB bn.)	174	238	91	194

Table J.3: Tax Effects (RMB bn.) Relative to the Observed Policy: Threshold Robustness

	Lower Threshold			Higher Threshold		
	(1) No Policy	(2) Optimal	(3) Weighted	(1) No Policy	(2) Optimal	(3) Weighted
Δ Consumer Surplus	530	-612	-270	610	-637	-473
Δ Producer Surplus	-236	-632	-557	-349	-643	-621
Δ Tax Revenue	-13	1278	974	-15	1259	1129
Δ Saved Pollution Damages	-290	257	125	-348	263	203
ΔW	-9	291	271	-101	242	237

Notes: The table reports changes in 2017 relative to the observed policy. Saved pollution damages include local air-pollution health benefits and water depletion, and exclude carbon. The tax levels in columns (1), (2), and (3) are 0, 420, and 293 RMB/ton under the lower-threshold specification, and 0, 415, and 357 RMB/ton under the higher-threshold specification.

Table J.4: Policy Weights That Rationalize Production Restrictions Over Taxes: Threshold Robustness

w_{damage}	1.00	1.25	1.50	1.75	2.00
Lower Threshold: $w_{\text{PS}} \geq$	1.486	1.563	1.671	1.801	1.950
Higher Threshold: $w_{\text{PS}} \geq$	1.382	1.481	1.607	1.753	1.914

Notes: The best-tax levels corresponding to the columns are Lower Threshold: 300, 400, 475, 575, 650; Higher Threshold: 350, 450, 550, 625, 700 RMB/ton.

Figure J.8: Estimated Aggregate Supply: Threshold Robustness

Lower Threshold | Higher Threshold

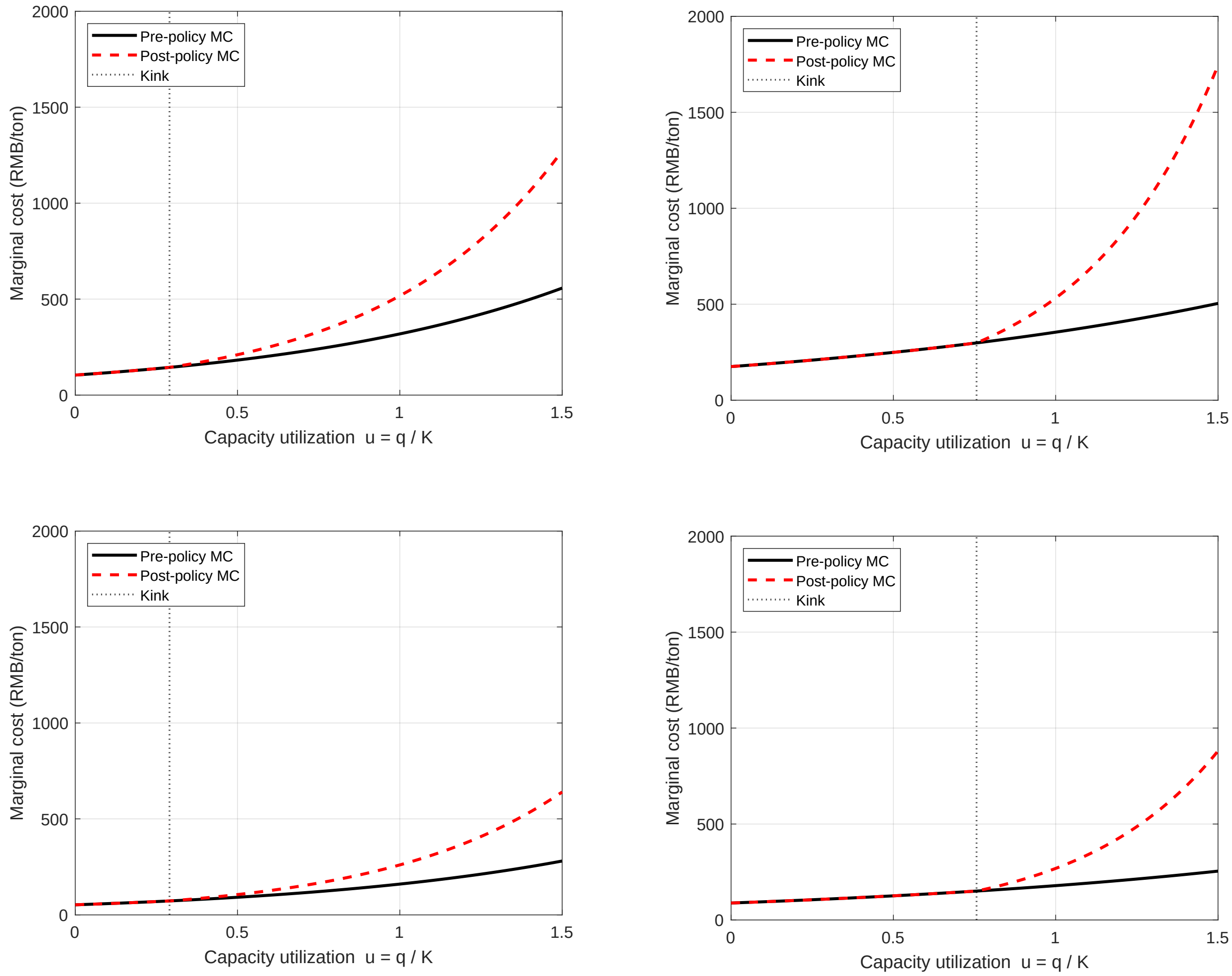


Figure J.9: Prices and Production With and Without the Policy: Threshold Robustness

Lower Threshold

Higher Threshold

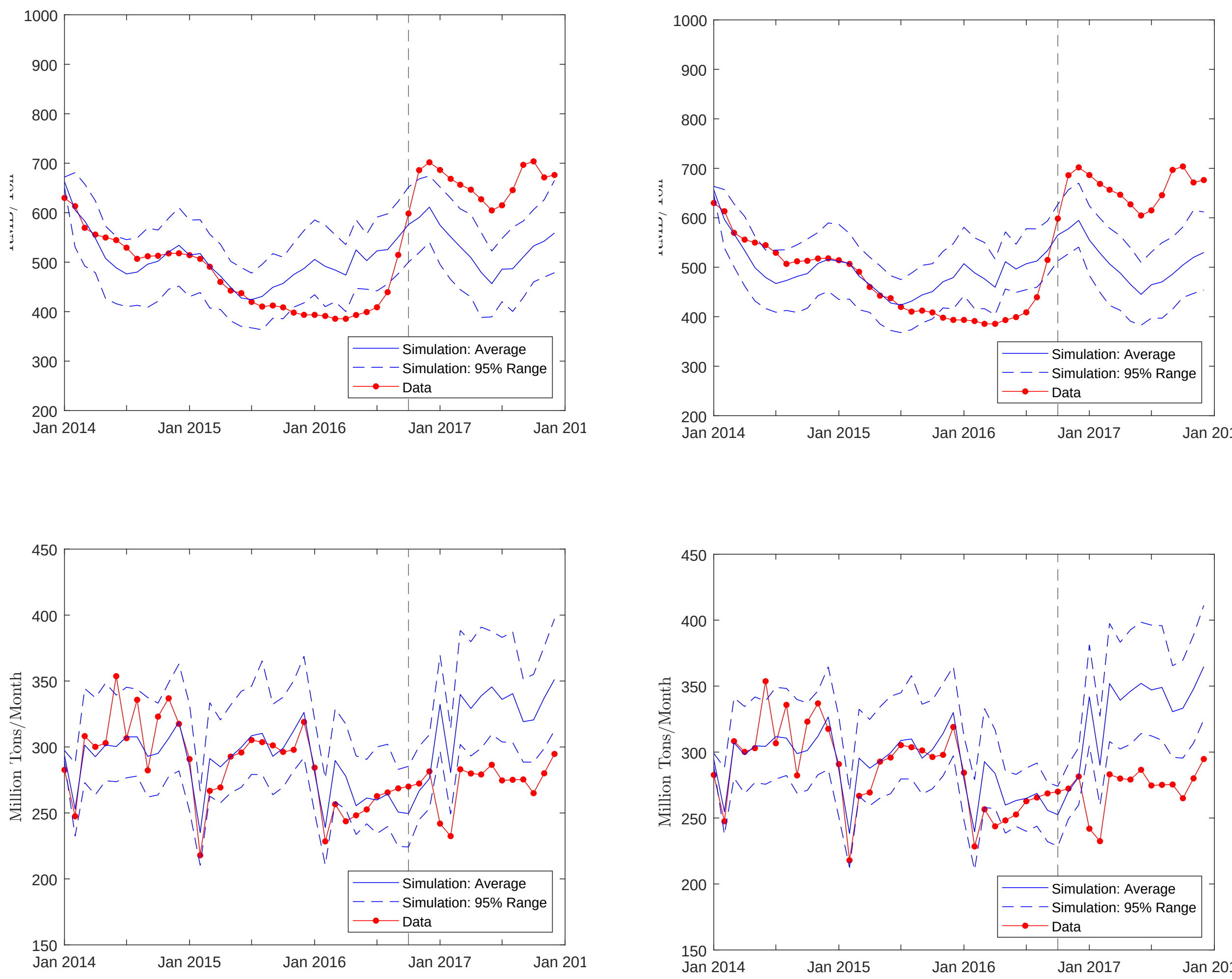


Figure J.10: Quantity CDFs: Production Restriction and Optimal Tax: Threshold Robustness

Lower Threshold | Higher Threshold

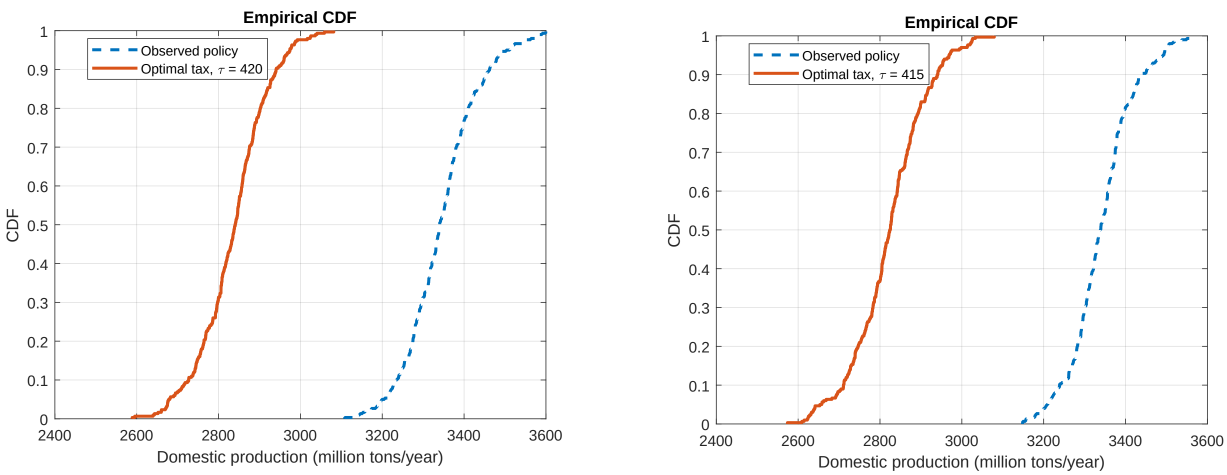


Figure J.11: Compare Estimated Average Costs with Unit Costs from Financial Reports: Threshold Robustness

Lower Threshold | Higher Threshold

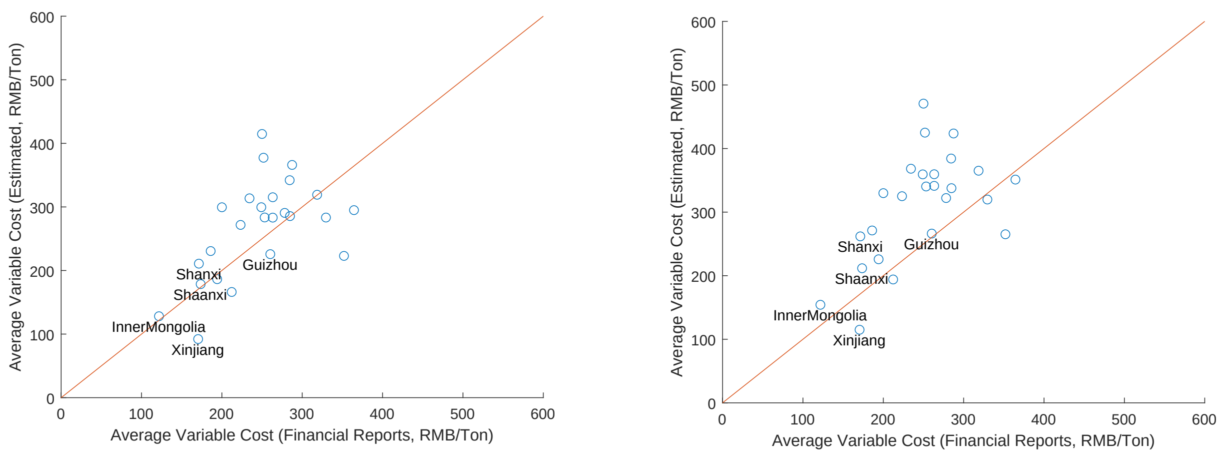


Figure J.12: Welfare Breakdown: Threshold Robustness